

Power Analysis for Confirmatory Factor Analysis: A Practical Tutorial

Analytical and Simulation-Based Approaches

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This notebook presents a pedagogical tutorial on statistical power analysis in Confirmatory Factor Analysis (CFA). Proper planning of sample sizes requires assessing the power to detect overall model misfit by analytical approach or simulation-based approach. Standard analytical tools typically rely on asymptotic normal theory, which may break down when confronted with non-normality or missing data. This tutorial demonstrates both analytical strategies (using `semPower` and `semTools`) and Monte Carlo simulations (using `simsem`) side-by-side using the WHOQOL-BREF 4-factor model as a unifying case study. We illustrate how simulation can complement analytical calculations, allowing researchers to evaluate parameter-level power, coverage, and relative bias under realistic scenarios involving varying degrees of model misspecification, missing data, and continuous non-normal indicators. Ultimately, we argue that analytical methods are ideal for initial planning, while simulations constitute the gold standard for robust confirmation.

1 Introduction

Statistical power in Confirmatory Factor Analysis (CFA) reflects the probability of correctly rejecting an incorrectly specified model, or detecting a parameter of interest when it is truly non-zero in the population (MacCallum et al., 1996). Power analysis is critical when planning a CFA study; however, researchers often rely on arbitrary rules of thumb mapping required sample sizes to the number of measured variables or parameters, rather than formal prospective calculations.

There are two main paradigms to power analysis in CFA:

1. **Analytical methods:** Rely on asymptotic statistical theory (e.g., the non-central χ^2 distribution) to calculate the required sample size for detecting global model misfit based on established fit indices like the RMSEA (Moshagen & Bader, 2024).

2. **Simulation-based methods:** Use Monte Carlo data generation to repeatedly establish how the model behaves under assumed population models, allowing the estimation of power, bias, and coverage for specific parameters or overall fit indices across varying sample sizes (Pornprasertmanit et al., 2021).

This document provides a comprehensive tutorial covering both analytical methods using the `semPower` and `semTools` packages, and Monte Carlo methods using the `simsem` package.

Reproducibility Note

For local development and computational convenience across this pedagogical document, simulations use `N_REP = 50`. For production-quality applied research, you must increase this to `N_REP = 1,000`. The results throughout this tutorial will thus show more sampling variability than you would expect in a final study. All analytical and simulation parameters are centrally defined in `helpers.R` and `models.R`.

2 The Running Example: WHOQOL-BREF Population CFA Model

To anchor our power analyses, we use the WHOQOL-BREF measurement instrument. Meaningful power analysis requires the definition of a population model—the “true” state of the world from which we believe our data are generated. By simulating from a meta-analytically derived population model rather than a purely arbitrary one, researchers can ensure that tests reflect realistic boundaries (Lin & Yao, 2022).

We specify three population variations to perform sensitivity analyses:

1. **Meta-analytic:** Loadings and residual variances closely matching Lin & Yao (2022).
2. **Naïve:** A pessimistic baseline where all factor loadings are weak ($\lambda = 0.5$).
3. **Optimistic:** A best-case scenario with stronger factor loadings ($\lambda = 0.7$).

2.1 Population model

The primary population model uses factor loadings derived from the meta-analysis by Lin and Yao (2022). Four factors: Psychological (6 items), Physical (7 items), Social (3 items), Environment (8 items). Interfactor correlations are set at the upper bound (0.3) reported in the meta-analysis (range: 0.08–0.30). Residual variances follow the formula $1 - \lambda^2$, except for Q3 and Q4 which share a correlated residual (0.3).

```
# Factor loadings from Lin and Yao (2022) meta-analysis
psycho =~ 0.92*Q5_P + 0.81*Q6_P + 0.94*Q7_P +
```

```

        0.73*Q11_P + 0.75*Q19_P + 0.63*Q26_P
physical =~ 0.84*Q3_F + 0.83*Q4_F + 0.63*Q10_F +
        0.71*Q15_F + 0.68*Q16_F + 0.89*Q17_F + 0.61*Q18_F
social =~ 0.94*Q20_S + 0.89*Q21_S + 0.78*Q22_S
environment =~ 0.37*Q8_A + 0.49*Q9_A + 0.79*Q12_A + 0.81*Q13_A +
        0.77*Q14_A + 0.75*Q23_A + 0.64*Q24_A + 0.73*Q25_A

# Correlated residual between Q3 and Q4
Q4_F ~~ 0.3*Q3_F

# Factor variances fixed to 1; interfactor correlations = 0.3
psycho ~~ 1*psycho + 0.3*physical + 0.3*social + 0.3*environment
physical ~~ 1*physical + 0.3*social + 0.3*environment
social ~~ 1*social + 0.3*environment
environment ~~ 1*environment

# Residual variances = 1 - lambda^2
Q5_P ~~ 0.15*Q5_P
Q6_P ~~ 0.34*Q6_P
Q7_P ~~ 0.12*Q7_P
Q11_P ~~ 0.47*Q11_P
Q19_P ~~ 0.44*Q19_P
Q26_P ~~ 0.60*Q26_P
Q10_F ~~ 0.60*Q10_F
Q15_F ~~ 0.50*Q15_F
Q16_F ~~ 0.54*Q16_F
Q17_F ~~ 0.21*Q17_F
Q18_F ~~ 0.63*Q18_F
Q20_S ~~ 0.12*Q20_S
Q21_S ~~ 0.21*Q21_S
Q22_S ~~ 0.39*Q22_S
Q8_A ~~ 0.86*Q8_A
Q9_A ~~ 0.76*Q9_A
Q12_A ~~ 0.38*Q12_A
Q13_A ~~ 0.34*Q13_A
Q14_A ~~ 0.41*Q14_A
Q23_A ~~ 0.44*Q23_A
Q24_A ~~ 0.59*Q24_A
Q25_A ~~ 0.47*Q25_A

```

Source: [Population Models](#)

2.2 Naive model

All factor loadings set to 0.5 (minimum acceptable for CFA), residual variances = 0.75 (i.e., $1 - 0.5^2$), and small interfactor correlations (0.3). This represents a **pessimistic** scenario.

```
# All loadings = 0.5
psycho =~ 0.5*Q5_P + 0.5*Q6_P + 0.5*Q7_P +
          0.5*Q11_P + 0.5*Q19_P + 0.5*Q26_P
physical =~ 0.5*Q3_F + 0.5*Q4_F + 0.5*Q10_F +
            0.5*Q15_F + 0.5*Q16_F + 0.5*Q17_F + 0.5*Q18_F
social =~ 0.5*Q20_S + 0.5*Q21_S + 0.5*Q22_S
environment =~ 0.5*Q8_A + 0.5*Q9_A + 0.5*Q12_A + 0.5*Q13_A +
               0.5*Q14_A + 0.5*Q23_A + 0.5*Q24_A + 0.5*Q25_A

Q4_F ~~ 0.3*Q3_F

psycho ~~ 1*psycho + 0.3*physical + 0.3*social + 0.3*environment
physical ~~ 1*physical + 0.3*social + 0.3*environment
social ~~ 1*social + 0.3*environment
environment ~~ 1*environment

# Residual variances = 1 - 0.5^2 = 0.75
Q5_P ~~ 0.75*Q5_P
Q6_P ~~ 0.75*Q6_P
Q7_P ~~ 0.75*Q7_P
Q11_P ~~ 0.75*Q11_P
Q19_P ~~ 0.75*Q19_P
Q26_P ~~ 0.75*Q26_P
Q3_F ~~ 0.75*Q3_F
Q4_F ~~ 0.75*Q4_F
Q10_F ~~ 0.75*Q10_F
Q15_F ~~ 0.75*Q15_F
Q16_F ~~ 0.75*Q16_F
Q17_F ~~ 0.75*Q17_F
Q18_F ~~ 0.75*Q18_F
Q20_S ~~ 0.75*Q20_S
Q21_S ~~ 0.75*Q21_S
Q22_S ~~ 0.75*Q22_S
Q8_A ~~ 0.75*Q8_A
Q9_A ~~ 0.75*Q9_A
Q12_A ~~ 0.75*Q12_A
```

```

Q13_A ~~ 0.75*Q13_A
Q14_A ~~ 0.75*Q14_A
Q23_A ~~ 0.75*Q23_A
Q24_A ~~ 0.75*Q24_A
Q25_A ~~ 0.75*Q25_A

```

Source: [Population Models](#)

2.3 Optimistic model

All factor loadings set to 0.7 (strong loadings), residual variances = 0.51 ($1 - 0.7^2$), and moderate interfactor correlations (0.5). This represents an **best-case** scenario.

```

# All loadings = 0.7
psycho =~ 0.7*Q5_P + 0.7*Q6_P + 0.7*Q7_P +
          0.7*Q11_P + 0.7*Q19_P + 0.7*Q26_P
physical =~ 0.7*Q3_F + 0.7*Q4_F + 0.7*Q10_F +
            0.7*Q15_F + 0.7*Q16_F + 0.7*Q17_F + 0.7*Q18_F
social =~ 0.7*Q20_S + 0.7*Q21_S + 0.7*Q22_S
environment =~ 0.7*Q8_A + 0.7*Q9_A + 0.7*Q12_A + 0.7*Q13_A +
              0.7*Q14_A + 0.7*Q23_A + 0.7*Q24_A + 0.7*Q25_A

Q4_F ~~ 0.5*Q3_F

psycho ~~ 1*psycho + 0.5*physical + 0.5*social + 0.5*environment
physical ~~ 1*physical + 0.5*social + 0.5*environment
social ~~ 1*social + 0.5*environment
environment ~~ 1*environment

# Residual variances = 1 - 0.7^2 = 0.51
Q5_P ~~ 0.51*Q5_P
Q6_P ~~ 0.51*Q6_P
Q7_P ~~ 0.51*Q7_P
Q11_P ~~ 0.51*Q11_P
Q19_P ~~ 0.51*Q19_P
Q26_P ~~ 0.51*Q26_P
Q3_F ~~ 0.51*Q3_F
Q4_F ~~ 0.51*Q4_F
Q10_F ~~ 0.51*Q10_F
Q15_F ~~ 0.51*Q15_F

```

```

Q16_F ~~ 0.51*Q16_F
Q17_F ~~ 0.51*Q17_F
Q18_F ~~ 0.51*Q18_F
Q20_S ~~ 0.51*Q20_S
Q21_S ~~ 0.51*Q21_S
Q22_S ~~ 0.51*Q22_S
Q8_A  ~~ 0.51*Q8_A
Q9_A  ~~ 0.51*Q9_A
Q12_A ~~ 0.51*Q12_A
Q13_A ~~ 0.51*Q13_A
Q14_A ~~ 0.51*Q14_A
Q23_A ~~ 0.51*Q23_A
Q24_A ~~ 0.51*Q24_A
Q25_A ~~ 0.51*Q25_A

```

Source: [Population Models](#)

2.4 Comparison table

Below is a summary of the three population models.

Table 1: Summary of population model variants used for sensitivity analysis

Model	Loading Range	Mean Loading	Factor Correlations	Residual Range	Q3–Q4 Covariance
Meta-analytic (popModel)	0.37–0.94	0.74	0.30	0.12–0.86	0.30
Naive (naiveModel)	0.50	0.50	0.30	0.75	0.30
Optimistic (optModel)	0.70	0.70	0.50	0.51	0.50

Source: [Population Models](#)

2.5 Analysis model

This is the model we will fit to data. All parameters are freely estimated (no fixed values), matching the structure of the population models.

```

psycho =~ Q5_P + Q6_P + Q7_P + Q11_P + Q19_P + Q26_P
physical =~ Q3_F + Q4_F + Q10_F + Q15_F + Q16_F + Q17_F + Q18_F
social =~ Q20_S + Q21_S + Q22_S
environment =~ Q8_A + Q9_A + Q12_A + Q13_A + Q14_A + Q23_A + Q24_A + Q25_A
Q4_F ~~ Q3_F

```

Source: [Population Models](#)

3 Analytical Power Analysis

Analytical power usually asks: “How large must N be to reject the hypothesis that the model fits perfectly (or adequately) when the true misfit in the population is of a certain magnitude?” A common formulation sets the alternative hypothesis to $RMSEA = 0.05$ (MacCallum et al., 1996).

3.1 Model Degrees of Freedom

Analytical calculations depend crucially on model degrees of freedom (df), the difference between the number of available pieces of information (unique variances/covariances) and the number of freely estimated parameters. Knowing this allows tools to parameterize the non-central χ^2 distribution. This calculation is necessary for both a priori and post hoc power analyses in analytical methods for global fit.

Model degrees of freedom (via `semPower`): 249

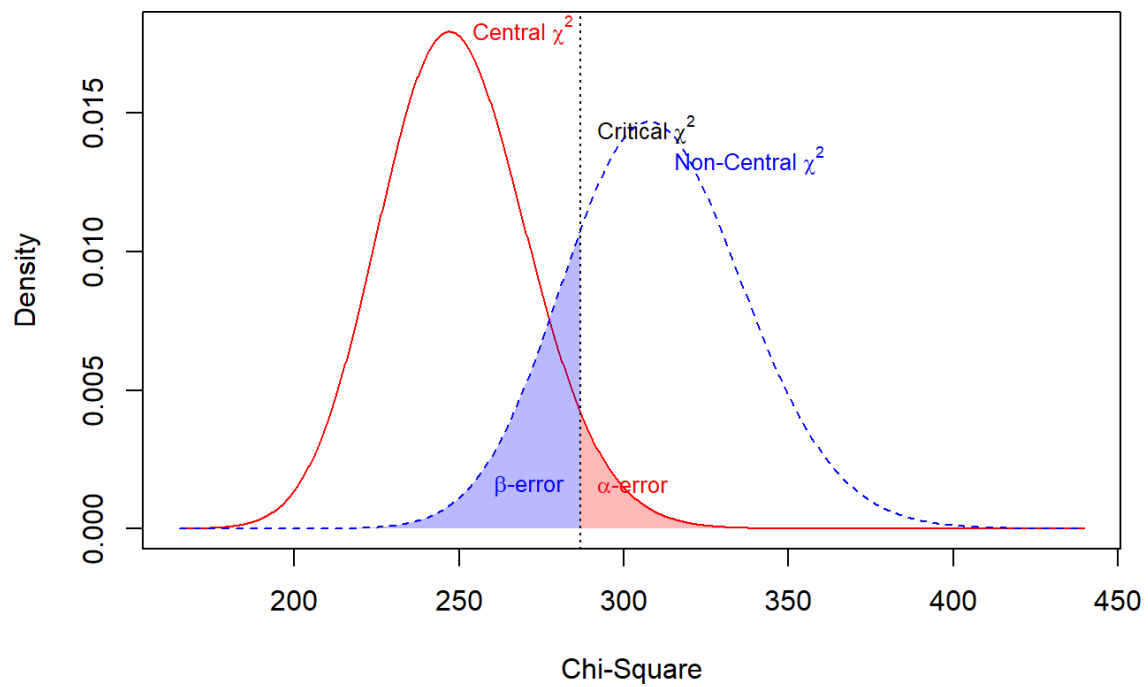
Source: [Analytical Power Analysis](#)

3.2 A Priori: How Many Observations?

`semPower`: A priori power analysis

F0	0.622500
RMSEA	0.050000
Mc	0.732531
GFI	0.950683
AGFI	0.940582

df	249
Required Num Observations	99
Critical Chi-Square	286.8077
NCP	61.00500
Alpha	0.050000
Beta	0.199217
Power (1 - Beta)	0.800783
Implied Alpha/Beta Ratio	0.250983

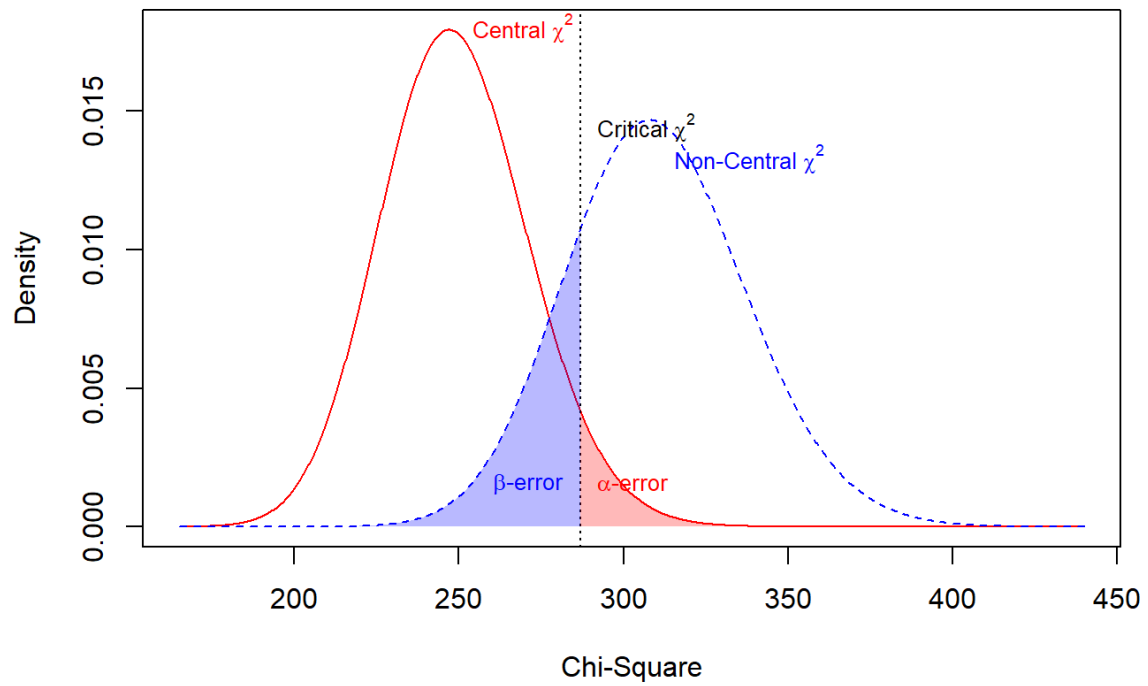


Source: [Analytical Power Analysis](#)

semPower: A priori power analysis

F0	0.285805
RMSEA	0.033879
Mc	0.866838

df	249
Required Num Observations	215
Critical Chi-Square	286.8077
NCP	61.16236
Alpha	0.050000
Beta	0.197630
Power (1 - Beta)	0.802370
Implied Alpha/Beta Ratio	0.252998



Source: [Analytical Power Analysis](#)

3.3 Power Curves

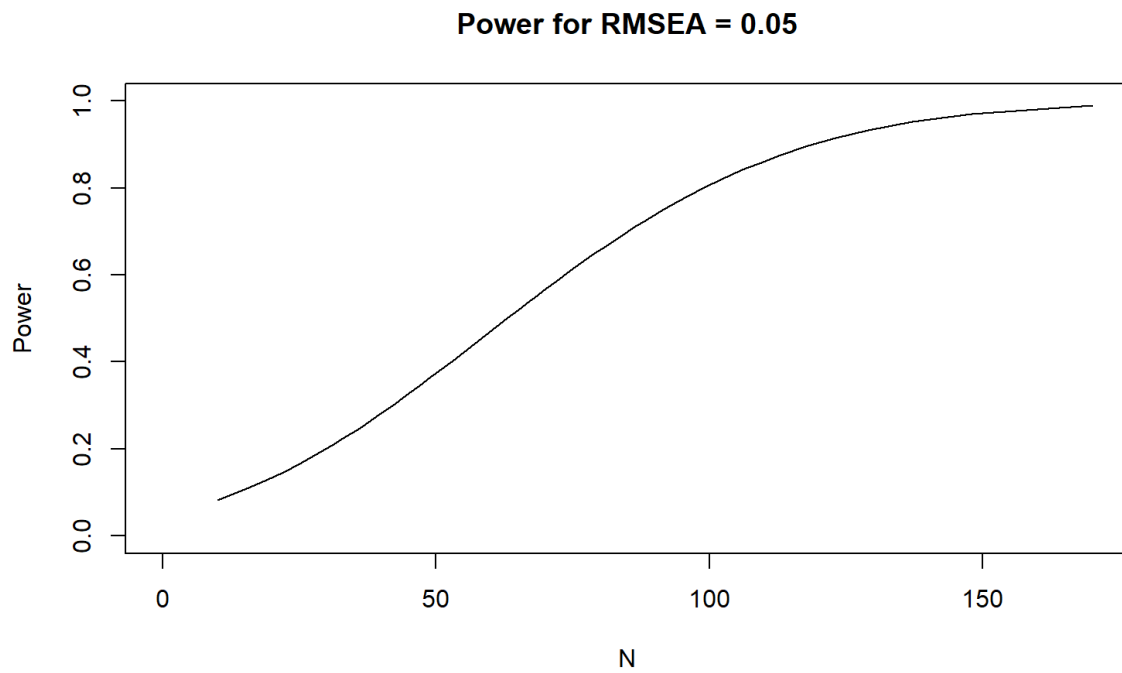


Figure 1: Power as a function of sample size for RMSEA = 0.05

Source: [Analytical Power Analysis](#)

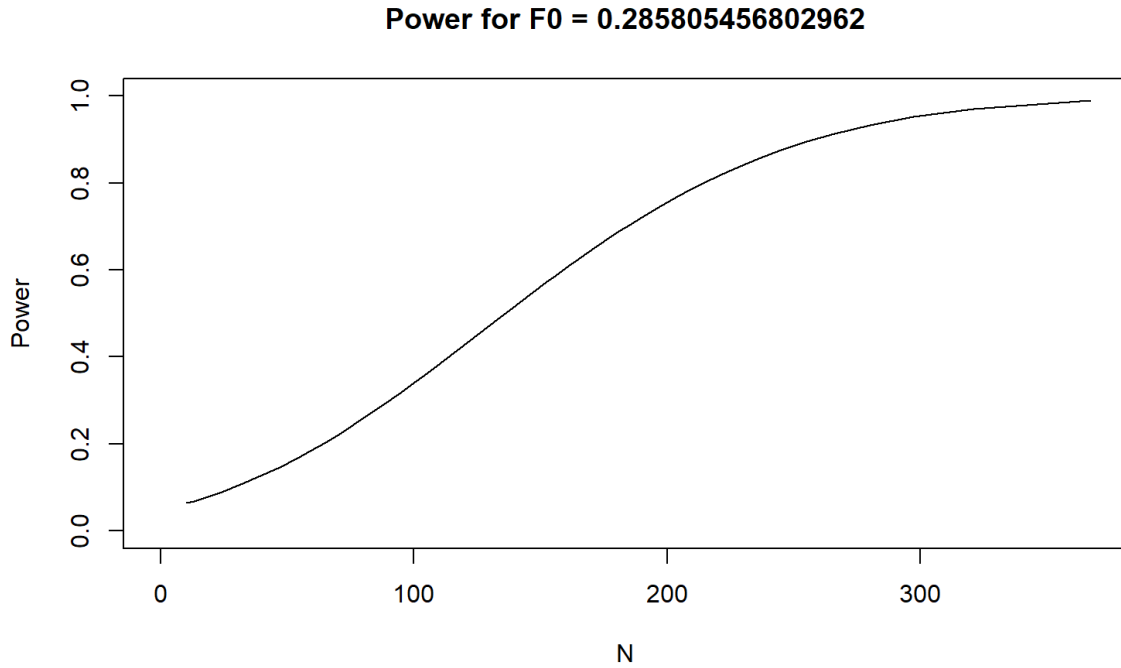


Figure 2: Power as a function of sample size for the population model (popModel)

Source: [Analytical Power Analysis](#)

Table 2: Summary of analytical power analysis results

Method	N Required	Power	Implied RMSEA	Implied Fmin	df
Model-free a priori (RMSEA .05)	99	0.8008	0.0500	0.6225	249
Model-based a priori (Population Model)	215	0.8024	0.0339	0.2858	249

Source: [Analytical Power Analysis](#)

! Limitations of the Analytical Approach

The RMSEA/ χ^2 -based analytical method classically assumes continuous, multivariate normal data and Maximum Likelihood estimation. It can yield severely biased optimal sample sizes when distributions are non-normal, data are ordinal, or missingness induces

efficiency losses. Furthermore, it measures power regarding *global misfit*, telling us nothing about whether N is adequate to detect a specific weak loading or structural path.

4 Monte Carlo Simulation: Ideal Conditions

Given the limitations above, Monte Carlo simulation stands as a highly flexible strategy. We generate many synthetic datasets from the defined `popModel`, apply our `analyzeModel`, and record the parameter estimates and standard errors.

Under strictly ideal conditions (multivariate normal continuous data, no missing values), simulation results should match analytical calculations, validating both implementations. Notice that varying sample sizes gives us empirical power curves.

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	314.747	5839.741	6005.252	0.040	0.968	0.964	0.068
2	325	297.720	18610.406	18811.685	0.031	0.976	0.973	0.056
3	550	280.694	31381.072	31618.118	0.022	0.984	0.983	0.045
4	775	263.668	44151.738	44424.551	0.012	0.992	0.992	0.033
5	1000	246.641	56922.404	57230.984	0.003	1.000	1.002	0.022

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5_P	0.929	0.043	0.038	
psycho=~Q6_P	0.828	0.050	0.041	
psycho=~Q7_P	0.944	0.039	0.038	
psycho=~Q11_P	0.724	0.037	0.043	
psycho=~Q19_P	0.765	0.048	0.043	
psycho=~Q26_P	0.656	0.046	0.045	
physical=~Q3_F	0.862	0.071	0.060	
physical=~Q4_F	0.840	0.071	0.060	
physical=~Q10_F	0.640	0.046	0.046	
physical=~Q15_F	0.724	0.051	0.045	
physical=~Q16_F	0.692	0.036	0.046	
physical=~Q17_F	0.891	0.034	0.041	
physical=~Q18_F	0.608	0.042	0.047	
social=~Q20_S	0.928	0.044	0.038	

social=~Q21_S	0.885	0.048	0.039
social=~Q22_S	0.773	0.034	0.042
environment=~Q8_A	0.383	0.048	0.049
environment=~Q9_A	0.497	0.035	0.047
environment=~Q12_A	0.797	0.035	0.042
environment=~Q13_A	0.819	0.036	0.042
environment=~Q14_A	0.773	0.050	0.043
environment=~Q23_A	0.768	0.042	0.044
environment=~Q24_A	0.638	0.068	0.046
environment=~Q25_A	0.717	0.058	0.043
Q3_F~~Q4_F	0.296	0.059	0.057
psycho~~physical	0.306	0.049	0.049
psycho~~social	0.287	0.058	0.048
psycho~~environment	0.310	0.061	0.048
physical~~social	0.286	0.054	0.050
physical~~environment	0.314	0.062	0.050
social~~environment	0.307	0.039	0.049
Q5_P~~Q5_P	0.142	0.018	0.015
Q6_P~~Q6_P	0.337	0.023	0.026
Q7_P~~Q7_P	0.127	0.017	0.015
Q11_P~~Q11_P	0.471	0.024	0.035
Q19_P~~Q19_P	0.441	0.043	0.033
Q26_P~~Q26_P	0.603	0.050	0.044
Q10_F~~Q10_F	0.595	0.056	0.045
Q15_F~~Q15_F	0.503	0.030	0.041
Q16_F~~Q16_F	0.556	0.046	0.044
Q17_F~~Q17_F	0.211	0.026	0.027
Q18_F~~Q18_F	0.637	0.043	0.047
Q20_S~~Q20_S	0.117	0.022	0.023
Q21_S~~Q21_S	0.199	0.028	0.024
Q22_S~~Q22_S	0.387	0.034	0.031
Q8_A~~Q8_A	0.845	0.059	0.059
Q9_A~~Q9_A	0.739	0.066	0.052
Q12_A~~Q12_A	0.375	0.032	0.032
Q13_A~~Q13_A	0.328	0.047	0.030
Q14_A~~Q14_A	0.404	0.037	0.033
Q23_A~~Q23_A	0.434	0.039	0.036
Q24_A~~Q24_A	0.606	0.046	0.046
Q25_A~~Q25_A	0.466	0.035	0.037
Q3_F~~Q3_F	0.984	0.075	0.076
Q4_F~~Q4_F	1.006	0.092	0.076

	Power (Not equal 0)	Std Est	Std Est	SD	Ave SE
psycho=~Q5_P	1.000	0.926	0.010		0.009

psycho=~Q6_P	1.000	0.818	0.020	0.017
psycho=~Q7_P	1.000	0.935	0.011	0.009
psycho=~Q11_P	1.000	0.725	0.020	0.025
psycho=~Q19_P	1.000	0.755	0.019	0.023
psycho=~Q26_P	1.000	0.645	0.030	0.030
physical=~Q3_F	1.000	0.654	0.036	0.031
physical=~Q4_F	1.000	0.641	0.039	0.032
physical=~Q10_F	1.000	0.638	0.039	0.032
physical=~Q15_F	1.000	0.713	0.029	0.028
physical=~Q16_F	1.000	0.680	0.025	0.030
physical=~Q17_F	1.000	0.889	0.015	0.017
physical=~Q18_F	1.000	0.605	0.032	0.034
social=~Q20_S	1.000	0.938	0.013	0.013
social=~Q21_S	1.000	0.892	0.019	0.015
social=~Q22_S	1.000	0.779	0.023	0.022
environment=~Q8_A	1.000	0.384	0.041	0.044
environment=~Q9_A	1.000	0.501	0.024	0.039
environment=~Q12_A	1.000	0.793	0.018	0.022
environment=~Q13_A	1.000	0.819	0.023	0.020
environment=~Q14_A	1.000	0.772	0.027	0.023
environment=~Q23_A	1.000	0.759	0.017	0.024
environment=~Q24_A	1.000	0.632	0.049	0.033
environment=~Q25_A	1.000	0.723	0.034	0.027
Q3_F~~Q4_F	1.000	0.297	0.050	0.049
psycho~~physical	1.000	0.306	0.049	0.049
psycho~~social	0.947	0.287	0.058	0.048
psycho~~environment	1.000	0.310	0.061	0.048
physical~~social	1.000	0.286	0.054	0.050
physical~~environment	1.000	0.314	0.062	0.050
social~~environment	1.000	0.307	0.039	0.049
Q5_P~~Q5_P	1.000	0.142	0.018	0.017
Q6_P~~Q6_P	1.000	0.331	0.032	0.029
Q7_P~~Q7_P	1.000	0.125	0.020	0.016
Q11_P~~Q11_P	1.000	0.473	0.029	0.036
Q19_P~~Q19_P	1.000	0.430	0.028	0.034
Q26_P~~Q26_P	1.000	0.583	0.039	0.038
Q10_F~~Q10_F	1.000	0.592	0.049	0.041
Q15_F~~Q15_F	1.000	0.491	0.040	0.039
Q16_F~~Q16_F	1.000	0.537	0.033	0.040
Q17_F~~Q17_F	1.000	0.210	0.027	0.029
Q18_F~~Q18_F	1.000	0.633	0.039	0.041
Q20_S~~Q20_S	1.000	0.120	0.025	0.025
Q21_S~~Q21_S	1.000	0.203	0.034	0.027

Q22_S~~Q22_S	1.000	0.393	0.035	0.034
Q8_A~~Q8_A	1.000	0.851	0.033	0.034
Q9_A~~Q9_A	1.000	0.749	0.024	0.040
Q12_A~~Q12_A	1.000	0.371	0.029	0.034
Q13_A~~Q13_A	1.000	0.328	0.038	0.032
Q14_A~~Q14_A	1.000	0.404	0.043	0.035
Q23_A~~Q23_A	1.000	0.424	0.026	0.036
Q24_A~~Q24_A	1.000	0.599	0.057	0.040
Q25_A~~Q25_A	1.000	0.477	0.047	0.038
Q3_F~~Q3_F	1.000	0.570	0.047	0.041
Q4_F~~Q4_F	1.000	0.587	0.050	0.041
	Average Param	Average Bias	Coverage	r_coef.n r_se.n
psycho=~Q5_P	0.92	0.009	0.947	-0.384 -0.883
psycho=~Q6_P	0.81	0.018	0.947	-0.545 -0.881
psycho=~Q7_P	0.94	0.004	0.947	-0.203 -0.897
psycho=~Q11_P	0.73	-0.006	0.947	-0.067 -0.892
psycho=~Q19_P	0.75	0.015	0.947	-0.138 -0.886
psycho=~Q26_P	0.63	0.026	0.947	-0.290 -0.886
physical=~Q3_F	0.84	0.022	0.895	-0.102 -0.886
physical=~Q4_F	0.83	0.010	0.895	-0.115 -0.913
physical=~Q10_F	0.63	0.010	0.895	0.012 -0.890
physical=~Q15_F	0.71	0.014	0.947	-0.030 -0.880
physical=~Q16_F	0.68	0.012	1.000	0.017 -0.873
physical=~Q17_F	0.89	0.001	1.000	-0.266 -0.894
physical=~Q18_F	0.61	-0.002	1.000	-0.131 -0.898
social=~Q20_S	0.94	-0.012	0.947	-0.015 -0.886
social=~Q21_S	0.89	-0.005	1.000	0.032 -0.885
social=~Q22_S	0.78	-0.007	0.895	-0.318 -0.877
environment=~Q8_A	0.37	0.013	0.895	0.031 -0.902
environment=~Q9_A	0.49	0.007	1.000	0.329 -0.930
environment=~Q12_A	0.79	0.007	1.000	0.225 -0.894
environment=~Q13_A	0.81	0.009	1.000	-0.054 -0.886
environment=~Q14_A	0.77	0.003	0.947	-0.294 -0.895
environment=~Q23_A	0.75	0.018	0.947	0.151 -0.890
environment=~Q24_A	0.64	-0.002	0.895	0.206 -0.902
environment=~Q25_A	0.73	-0.013	0.895	0.406 -0.911
Q3_F~~Q4_F	0.30	-0.004	1.000	0.142 -0.896
psycho~~physical	0.30	0.006	0.947	-0.312 -0.879
psycho~~social	0.30	-0.013	1.000	-0.008 -0.859
psycho~~environment	0.30	0.010	0.842	-0.201 -0.873
physical~~social	0.30	-0.014	1.000	-0.102 -0.875
physical~~environment	0.30	0.014	0.842	-0.419 -0.870
social~~environment	0.30	0.007	1.000	-0.311 -0.894

Q5_P~~Q5_P	0.15	-0.008	0.947	0.260	-0.858
Q6_P~~Q6_P	0.34	-0.003	1.000	0.225	-0.860
Q7_P~~Q7_P	0.12	0.007	0.895	-0.120	-0.872
Q11_P~~Q11_P	0.47	0.001	1.000	-0.279	-0.890
Q19_P~~Q19_P	0.44	0.001	0.895	-0.356	-0.847
Q26_P~~Q26_P	0.60	0.003	0.947	-0.116	-0.859
Q10_F~~Q10_F	0.60	-0.005	0.947	-0.024	-0.856
Q15_F~~Q15_F	0.50	0.003	1.000	-0.384	-0.867
Q16_F~~Q16_F	0.54	0.016	0.947	-0.270	-0.859
Q17_F~~Q17_F	0.21	0.001	1.000	0.110	-0.880
Q18_F~~Q18_F	0.63	0.007	0.947	0.231	-0.887
Q20_S~~Q20_S	0.12	-0.003	0.947	-0.479	-0.823
Q21_S~~Q21_S	0.21	-0.011	0.895	0.206	-0.817
Q22_S~~Q22_S	0.39	-0.003	0.947	0.175	-0.811
Q8_A~~Q8_A	0.86	-0.015	0.947	0.508	-0.896
Q9_A~~Q9_A	0.76	-0.021	0.947	0.441	-0.934
Q12_A~~Q12_A	0.38	-0.005	0.895	-0.016	-0.872
Q13_A~~Q13_A	0.34	-0.012	0.789	-0.380	-0.878
Q14_A~~Q14_A	0.41	-0.006	1.000	0.486	-0.903
Q23_A~~Q23_A	0.44	-0.006	1.000	-0.183	-0.879
Q24_A~~Q24_A	0.59	0.016	1.000	-0.431	-0.880
Q25_A~~Q25_A	0.47	-0.004	1.000	-0.179	-0.898
Q3_F~~Q3_F	1.00	-0.016	0.947	0.166	-0.875
Q4_F~~Q4_F	1.00	0.006	0.947	0.380	-0.898

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.355	-0.355	0.861	-0.708	-0.823	0.555	-0.354
aic	-0.355	1.000	1.000	-0.594	0.557	0.398	-0.867	1.000
bic	-0.355	1.000	1.000	-0.594	0.557	0.399	-0.867	1.000
rmsea	0.861	-0.594	-0.594	1.000	-0.937	-0.937	0.829	-0.594
cfi	-0.708	0.557	0.557	-0.937	1.000	0.955	-0.847	0.557
tli	-0.823	0.398	0.399	-0.937	0.955	1.000	-0.740	0.398
srmr	0.555	-0.867	-0.867	0.829	-0.847	-0.740	1.000	-0.866
n	-0.354	1.000	1.000	-0.594	0.557	0.398	-0.866	1.000

===== Replications =====

Number of replications = 19

Number of converged replications = 19

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Ideal Conditions](#)

--- Power at N = 300 ---

cfi	tli	rmsea	srmr
"< 0.48"	"< 0.5"	"< 0.48"	"< 0.46"

--- Power at N = 500 ---

cfi	tli	rmsea	srmr
"< 0.48"	"< 0.5"	"< 0.48"	"< 0.46"

--- Power at N = 1000 ---

cfi	tli	rmsea	srmr
"< 0.46"	"< 0.3"	"< 0.46"	"< 0.48"

Source: [Simulation — Ideal Conditions](#)

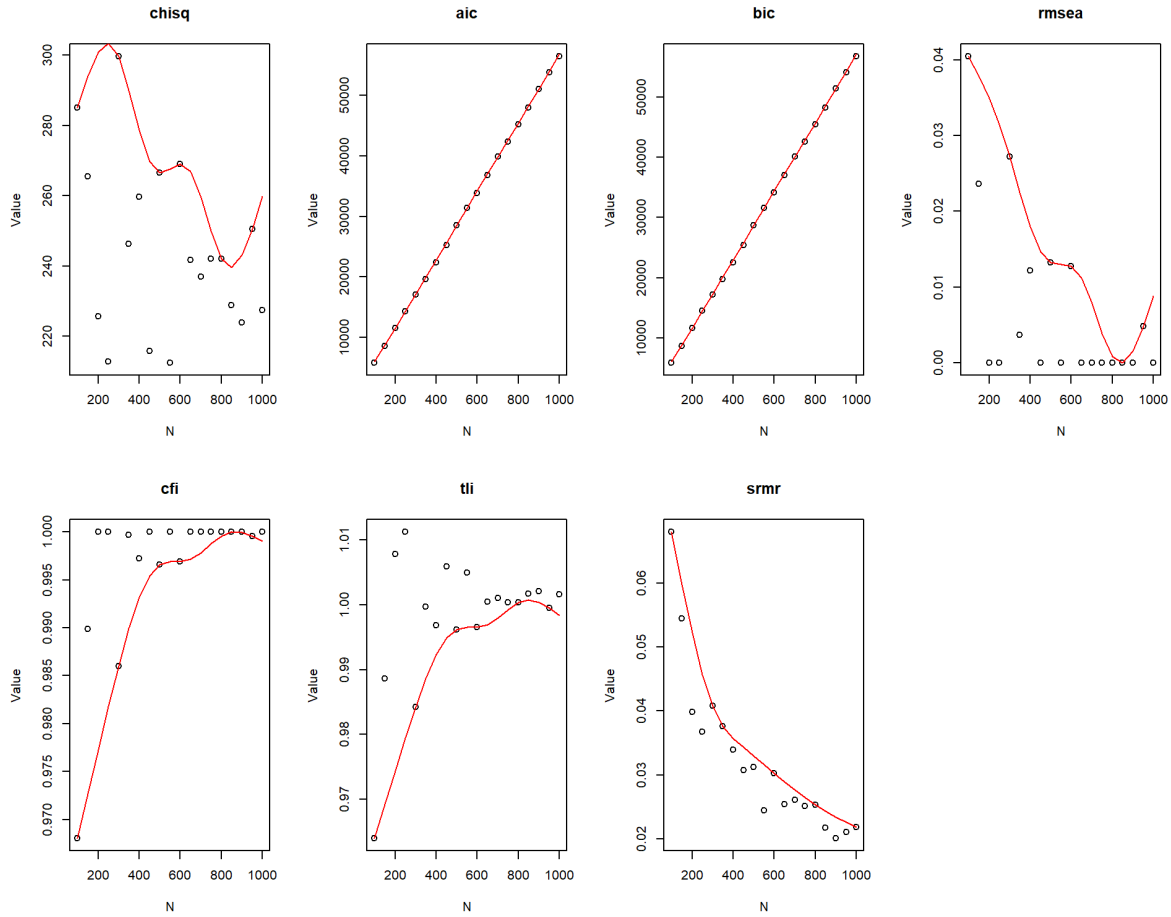


Figure 3: Sampling distributions of fit indices across varying N (ideal conditions)

Source: [Simulation — Ideal Conditions](#)

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Es- ti- mate	Es- ti- mate SD	Av- er- age SE	Power (Not equal 0)	Std Est	Std Est SD	Std Est SE	Av- er- age Par	Av- er- age Bias	Cov- er- age	Rel Bias	Std Bias	SE	Not Cove low	Av- er- age Cove Above	SD CI	SD CI	Width	Width
psy- cho=Q5_P	0.918	0.033	0.034	1	0.92	0.009	0.009	0.92	- 0.002	0.96	- 0.003	0.074		0.028	0.02	0.02	0.134	0.003	

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Es- ti- mate	Es- ti- mate	Av- er- age	Power (Not equal 0)	Std Est	Std SE	Av- er- age	Av- er- age	Cov- er- age	Rel Bias	Std Bias	SE Bias	Not Cove- low	Av- er- age	SD CI	SD CI	Width
psy- cho=~Q6_P	0.813	0.037	0.037	1	0.815	0.017	0.016	0.81	0.003	0.94	0.003	0.073	0.005	0.02	0.04	0.145	0.004
psy- cho=~Q7_P	0.940	0.032	0.034	1	0.940	0.008	0.008	0.94	0.000	0.96	0.000	0.007	0.056	0.02	0.02	0.133	0.004
psy- cho=~Q11_P	0.729	0.043	0.039	1	0.729	0.024	0.022	0.73	-	0.96	-	-	-	0.02	0.02	0.153	0.004
psy- cho=~Q19_P	0.751	0.039	0.039	1	0.750	0.022	0.021	0.75	0.001	0.96	0.001	0.020	-	0.02	0.02	0.152	0.004
psy- cho=~Q26_P	0.631	0.039	0.041	1	0.629	0.028	0.028	0.63	0.001	0.96	0.001	0.023	0.051	0.02	0.02	0.161	0.004
physi- cal=~Q3_F	0.834	0.054	0.055	1	0.834	0.029	0.030	0.84	-	0.90	-	-	0.016	0.02	0.08	0.215	0.006
physi- cal=~Q4_F	0.835	0.056	0.055	1	0.839	0.027	0.030	0.83	0.005	0.90	0.006	0.089	-	0.02	0.08	0.215	0.006
physi- cal=~Q10_F	0.636	0.040	0.042	1	0.634	0.029	0.030	0.63	0.006	0.98	0.010	0.148	0.041	0.02	0.00	0.165	0.005
physi- cal=~Q15_F	0.715	0.048	0.041	1	0.710	0.027	0.025	0.71	0.005	0.94	0.007	0.110	-	0.04	0.02	0.160	0.004
physi- cal=~Q16_F	0.686	0.037	0.042	1	0.681	0.027	0.027	0.68	0.006	0.98	0.009	0.163	0.125	0.00	0.02	0.163	0.004
physi- cal=~Q17_F	0.892	0.035	0.037	1	0.889	0.016	0.015	0.89	0.002	0.98	0.002	0.052	0.042	0.00	0.02	0.145	0.004
physi- cal=~Q18_F	0.609	0.047	0.042	1	0.610	0.031	0.031	0.61	-	0.96	-	-	-	0.02	0.02	0.166	0.005
so- cial=~Q20_S	0.935	0.039	0.035	1	0.936	0.013	0.012	0.94	-	0.94	-	-	-	0.00	0.06	0.138	0.003
so- cial=~Q21_S	0.883	0.033	0.036	1	0.887	0.014	0.014	0.89	-	0.96	-	-	0.079	0.04	0.00	0.142	0.003
so- cial=~Q22_S	0.777	0.037	0.038	1	0.781	0.017	0.020	0.78	-	0.98	-	-	0.036	0.00	0.02	0.150	0.004
envi- ron- ment=~Q8_A	0.366	0.040	0.045	1	0.368	0.036	0.041	0.37	-	0.98	-	-	0.141	0.00	0.02	0.178	0.005

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Es- ti- mate	Es- ti- mate	Av- er- age	Power (Not equal 0)	Std Est	Std Est	Av- er- age	Av- er- age	Cov- er- age	Rel Bias	Std Bias	SE	Rel Bias	Not Cove- low	Av- er- age	SD
envi- ron- ment=Q9_A	0.498	0.040	0.044	1	0.495	0.033	0.036	0.49	0.008	0.98	0.016	0.189	0.096	0.02	0.00	0.173
envi- ron- ment=Q12_A	0.780	0.041	0.039	1	0.784	0.023	0.021	0.79	-	0.96	-	-	-	0.00	0.04	0.151
envi- ron- ment=Q13_A	0.804	0.035	0.038	1	0.811	0.017	0.019	0.81	-	0.94	-	-	0.076	0.02	0.04	0.149
envi- ron- ment=Q14_A	0.766	0.038	0.039	1	0.767	0.022	0.022	0.77	-	0.96	-	-	0.038	0.00	0.04	0.153
envi- ron- ment=Q23_A	0.739	0.035	0.039	1	0.746	0.024	0.023	0.75	-	0.96	-	-	0.130	0.00	0.04	0.154
envi- ron- ment=Q24_A	0.634	0.044	0.042	1	0.636	0.028	0.029	0.64	-	0.96	-	-	0.02	0.02	0.163	0.005
envi- ron- ment=Q25_A	0.720	0.038	0.040	1	0.723	0.025	0.024	0.73	-	0.90	-	-	0.064	0.02	0.08	0.157
Q3_F~Q130	0.430	0.044	0.053	1	0.299	0.037	0.044	0.30	0.000	0.96	-	-	0.191	0.02	0.02	0.206
psy- cho~~phys- ical	0.305	0.051	0.044	1	0.305	0.051	0.044	0.30	0.005	0.90	0.017	0.103	-	0.06	0.04	0.174
psy- cho~~so- cial	0.297	0.047	0.044	1	0.297	0.047	0.044	0.30	-	0.96	-	-	-	0.02	0.02	0.171
psy- cho~~en- viron- ment	0.307	0.048	0.044	1	0.307	0.048	0.044	0.30	0.007	0.94	0.024	0.154	-	0.04	0.02	0.173

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Es- ti- mate	Es- ti- mate	Av- er- age	Power (Not equal 0)	Std Est	Std SE	Av- er- age	Av- er- age	Cov- er- age	Rel Bias	Std Bias	SE Bias	Not Cove- low	Av- er- age	SD CI	SD CI	Width
physi- cal~so- cial	0.301	0.043	0.045	1	0.301	0.043	0.045	0.30	0.001	0.96	0.003	0.023	0.054	0.04	0.00	0.177	0.005
physi- cal~en- viron- ment	0.311	0.052	0.046	1	0.311	0.052	0.046	0.30	0.011	0.94	0.036	0.209	- 0.121	0.06	0.00	0.179	0.006
so- cial~en- viron- ment	0.292	0.047	0.045	1	0.292	0.047	0.045	0.30	- 0.008	0.94	- 0.026	- 0.165	- 0.028	0.02	0.04	0.177	0.005
Q5_P~Q5_F	0.514	0.013	0.014	1	0.150	0.017	0.016	0.15	- 0.002	0.96	- 0.017	- 0.185	0.054	0.00	0.04	0.055	0.003
Q6_P~Q6_F	0.633	0.021	0.024	1	0.336	0.027	0.027	0.34	- 0.007	0.94	- 0.021	- 0.327	0.103	0.00	0.06	0.093	0.005
Q7_P~Q7_F	0.711	0.015	0.013	1	0.117	0.016	0.015	0.12	- 0.003	0.94	- 0.026	- 0.211	- 0.100	0.00	0.06	0.052	0.003
Q11_P~Q11_F	0.466	0.029	0.031	1	0.468	0.035	0.032	0.47	- 0.004	0.96	- 0.008	- 0.134	0.078	0.00	0.04	0.123	0.007
Q19_P~Q19_F	0.498	0.030	0.030	1	0.438	0.033	0.031	0.44	- 0.002	0.94	- 0.004	- 0.065	- 0.010	0.04	0.02	0.117	0.007
Q26_P~Q26_F	0.608	0.036	0.040	1	0.604	0.035	0.035	0.60	0.008	0.96	0.013	0.207	0.096	0.04	0.00	0.156	0.009
Q10_F~Q10_P	0.600	0.046	0.041	1	0.597	0.037	0.038	0.60	0.000	0.94	0.000	0.001	- 0.089	0.02	0.04	0.163	0.011
Q15_F~Q15_P	0.496	0.029	0.036	1	0.493	0.038	0.036	0.50	- 0.004	1.00	- 0.007	- 0.129	0.247	0.00	0.00	0.142	0.007
Q16_F~Q16_P	0.564	0.042	0.039	1	0.536	0.037	0.037	0.54	0.004	0.92	0.008	0.104	- 0.085	0.04	0.04	0.152	0.010
Q17_F~Q17_P	0.271	0.025	0.025	1	0.210	0.028	0.027	0.21	0.001	0.96	0.006	0.049	- 0.007	0.00	0.04	0.099	0.006
Q18_F~Q18_P	0.681	0.041	0.042	1	0.626	0.038	0.038	0.63	- 0.009	0.96	- 0.014	- 0.220	0.038	0.00	0.04	0.166	0.010
Q20_S~Q20_F	0.222	0.021	0.022	1	0.123	0.025	0.023	0.12	0.002	0.96	0.016	0.091	0.033	0.04	0.00	0.085	0.004

Table 3: Parameter estimates summary — Ideal conditions, fixed N = 500

	Es- ti- mate	Es- ti- mate	Av- er- age	Power (Not equal)	Std Est	Std Est	Av- er- age	Av- er- age	Cov- er- age	Rel Bias	Std Bias	SE	Not Cove- low	Av- er- age	SD
	age	SD	SE	0)	Est	SD	SE	Par	Bias	age	Bias	Bias	Bias	low	Width
Q21_S	0.210	0.024	0.022	1	0.212	0.025	0.025	0.21	0.000	0.98	-	-	-	0.00	0.02
Q22_S	0.234	0.027	0.028	1	0.389	0.027	0.031	0.39	-	0.96	-	-	0.04	0.04	0.11
Q8_A	0.855	0.053	0.055	1	0.864	0.026	0.030	0.86	-	0.96	-	-	0.05	0.04	0.21
Q9_A	0.976	0.057	0.050	1	0.754	0.033	0.036	0.76	0.000	0.88	0.000	0.002	-	0.04	0.08
Q12_A	0.321	0.029	0.030	1	0.385	0.036	0.032	0.38	0.001	0.96	0.002	0.026	0.026	0.04	0.11
Q13_A	0.337	0.029	0.027	1	0.342	0.028	0.030	0.34	-	0.88	-	-	-	0.04	0.08
Q14_A	0.409	0.031	0.031	1	0.412	0.034	0.033	0.41	-	0.92	-	-	-	0.02	0.06
Q23_A	0.436	0.037	0.032	1	0.444	0.035	0.034	0.44	-	0.92	-	-	-	0.00	0.08
Q24_A	0.591	0.036	0.041	1	0.595	0.036	0.037	0.59	0.001	0.96	0.001	0.024	0.140	0.02	0.15
Q25_A	0.472	0.036	0.034	1	0.477	0.035	0.035	0.47	0.002	0.96	0.004	0.049	-	0.00	0.04
Q3_F	0.898	0.066	0.070	1	0.589	0.036	0.038	1.00	-	0.96	-	-	0.05	0.00	0.04
Q4_F	0.407	0.057	0.070	1	0.591	0.035	0.038	1.00	0.007	0.98	0.007	0.130	0.221	0.02	0.00

Source: [Simulation — Ideal Conditions](#)

Table 4: Convergence check — Analytical vs. simulation N requirements

Approach	N for 80% Power	Notes
semPower a priori (RMSEA .05)	See Section 3.2	Model-free, RMSEA-based
semTools a priori (RMSEA .05)	See Section 3.2	Same underlying math
simsem (parameter power, max N)	From findPower() above	Continuous/ML, population model

Table 4: Convergence check — Analytical vs. simulation N requirements

Approach	N for 80% Power	Notes
simsem (fit power at SRMR < .06)	From getPowerFit() above	Continuous/ML, population model

Source: [Simulation — Ideal Conditions](#)

💡 Convergence Check

If the simulation-based N requirement closely matches the analytical estimate (or is slightly larger to account for sampling error), you have correctly operationalized the model in the simulation structure. This serves as a vital sanity check before introducing complex non-ideal conditions.

5 Monte Carlo Simulation: Realistic Scenarios

Most applied survey data violate the ideal assumptions. Simulating how analytical robustness deteriorates helps us adjust the planned sample sizes to offset the expected penalty.

5.1 Non-Normality

Applying continuous but non-normal (e.g., highly skewed or kurtotic) generation exposes the vulnerability of unadjusted normal-theory ML tests, emphasizing the necessity of robust estimators (e.g., MLR).

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	314.747	5839.741	6005.252	0.040	0.968	0.964	0.068
2	325	297.720	18610.406	18811.685	0.031	0.976	0.973	0.056
3	550	280.694	31381.072	31618.118	0.022	0.984	0.983	0.045
4	775	263.668	44151.738	44424.551	0.012	0.992	0.992	0.033
5	1000	246.641	56922.404	57230.984	0.003	1.000	1.002	0.022

===== Parameter Estimates and Standard Errors =====

Estimate Average Estimate SD Average SE

psycho=~Q5_P	0.929	0.043	0.038
psycho=~Q6_P	0.828	0.050	0.041
psycho=~Q7_P	0.944	0.039	0.038
psycho=~Q11_P	0.724	0.037	0.043
psycho=~Q19_P	0.765	0.048	0.043
psycho=~Q26_P	0.656	0.046	0.045
physical=~Q3_F	0.862	0.071	0.060
physical=~Q4_F	0.840	0.071	0.060
physical=~Q10_F	0.640	0.046	0.046
physical=~Q15_F	0.724	0.051	0.045
physical=~Q16_F	0.692	0.036	0.046
physical=~Q17_F	0.891	0.034	0.041
physical=~Q18_F	0.608	0.042	0.047
social=~Q20_S	0.928	0.044	0.038
social=~Q21_S	0.885	0.048	0.039
social=~Q22_S	0.773	0.034	0.042
environment=~Q8_A	0.383	0.048	0.049
environment=~Q9_A	0.497	0.035	0.047
environment=~Q12_A	0.797	0.035	0.042
environment=~Q13_A	0.819	0.036	0.042
environment=~Q14_A	0.773	0.050	0.043
environment=~Q23_A	0.768	0.042	0.044
environment=~Q24_A	0.638	0.068	0.046
environment=~Q25_A	0.717	0.058	0.043
Q3_F~~Q4_F	0.296	0.059	0.057
psycho~~physical	0.306	0.049	0.049
psycho~~social	0.287	0.058	0.048
psycho~~environment	0.310	0.061	0.048
physical~~social	0.286	0.054	0.050
physical~~environment	0.314	0.062	0.050
social~~environment	0.307	0.039	0.049
Q5_P~~Q5_P	0.142	0.018	0.015
Q6_P~~Q6_P	0.337	0.023	0.026
Q7_P~~Q7_P	0.127	0.017	0.015
Q11_P~~Q11_P	0.471	0.024	0.035
Q19_P~~Q19_P	0.441	0.043	0.033
Q26_P~~Q26_P	0.603	0.050	0.044
Q10_F~~Q10_F	0.595	0.056	0.045
Q15_F~~Q15_F	0.503	0.030	0.041
Q16_F~~Q16_F	0.556	0.046	0.044
Q17_F~~Q17_F	0.211	0.026	0.027
Q18_F~~Q18_F	0.637	0.043	0.047
Q20_S~~Q20_S	0.117	0.022	0.023

Q21_S~~Q21_S	0.199	0.028	0.024
Q22_S~~Q22_S	0.387	0.034	0.031
Q8_A~~Q8_A	0.845	0.059	0.059
Q9_A~~Q9_A	0.739	0.066	0.052
Q12_A~~Q12_A	0.375	0.032	0.032
Q13_A~~Q13_A	0.328	0.047	0.030
Q14_A~~Q14_A	0.404	0.037	0.033
Q23_A~~Q23_A	0.434	0.039	0.036
Q24_A~~Q24_A	0.606	0.046	0.046
Q25_A~~Q25_A	0.466	0.035	0.037
Q3_F~~Q3_F	0.984	0.075	0.076
Q4_F~~Q4_F	1.006	0.092	0.076

	Power (Not equal 0)	Std Est	Std Est SD	Std Ave SE
psycho=~Q5_P	1.000	0.926	0.010	0.009
psycho=~Q6_P	1.000	0.818	0.020	0.017
psycho=~Q7_P	1.000	0.935	0.011	0.009
psycho=~Q11_P	1.000	0.725	0.020	0.025
psycho=~Q19_P	1.000	0.755	0.019	0.023
psycho=~Q26_P	1.000	0.645	0.030	0.030
physical=~Q3_F	1.000	0.654	0.036	0.031
physical=~Q4_F	1.000	0.641	0.039	0.032
physical=~Q10_F	1.000	0.638	0.039	0.032
physical=~Q15_F	1.000	0.713	0.029	0.028
physical=~Q16_F	1.000	0.680	0.025	0.030
physical=~Q17_F	1.000	0.889	0.015	0.017
physical=~Q18_F	1.000	0.605	0.032	0.034
social=~Q20_S	1.000	0.938	0.013	0.013
social=~Q21_S	1.000	0.892	0.019	0.015
social=~Q22_S	1.000	0.779	0.023	0.022
environment=~Q8_A	1.000	0.384	0.041	0.044
environment=~Q9_A	1.000	0.501	0.024	0.039
environment=~Q12_A	1.000	0.793	0.018	0.022
environment=~Q13_A	1.000	0.819	0.023	0.020
environment=~Q14_A	1.000	0.772	0.027	0.023
environment=~Q23_A	1.000	0.759	0.017	0.024
environment=~Q24_A	1.000	0.632	0.049	0.033
environment=~Q25_A	1.000	0.723	0.034	0.027
Q3_F~~Q4_F	1.000	0.297	0.050	0.049
psycho~~physical	1.000	0.306	0.049	0.049
psycho~~social	0.947	0.287	0.058	0.048
psycho~~environment	1.000	0.310	0.061	0.048
physical~~social	1.000	0.286	0.054	0.050
physical~~environment	1.000	0.314	0.062	0.050

social~~environment	1.000	0.307	0.039	0.049
Q5_P~~Q5_P	1.000	0.142	0.018	0.017
Q6_P~~Q6_P	1.000	0.331	0.032	0.029
Q7_P~~Q7_P	1.000	0.125	0.020	0.016
Q11_P~~Q11_P	1.000	0.473	0.029	0.036
Q19_P~~Q19_P	1.000	0.430	0.028	0.034
Q26_P~~Q26_P	1.000	0.583	0.039	0.038
Q10_F~~Q10_F	1.000	0.592	0.049	0.041
Q15_F~~Q15_F	1.000	0.491	0.040	0.039
Q16_F~~Q16_F	1.000	0.537	0.033	0.040
Q17_F~~Q17_F	1.000	0.210	0.027	0.029
Q18_F~~Q18_F	1.000	0.633	0.039	0.041
Q20_S~~Q20_S	1.000	0.120	0.025	0.025
Q21_S~~Q21_S	1.000	0.203	0.034	0.027
Q22_S~~Q22_S	1.000	0.393	0.035	0.034
Q8_A~~Q8_A	1.000	0.851	0.033	0.034
Q9_A~~Q9_A	1.000	0.749	0.024	0.040
Q12_A~~Q12_A	1.000	0.371	0.029	0.034
Q13_A~~Q13_A	1.000	0.328	0.038	0.032
Q14_A~~Q14_A	1.000	0.404	0.043	0.035
Q23_A~~Q23_A	1.000	0.424	0.026	0.036
Q24_A~~Q24_A	1.000	0.599	0.057	0.040
Q25_A~~Q25_A	1.000	0.477	0.047	0.038
Q3_F~~Q3_F	1.000	0.570	0.047	0.041
Q4_F~~Q4_F	1.000	0.587	0.050	0.041
Average Param Average Bias Coverage r_coef.n r_se.n				
psycho=~Q5_P	0.92	0.009	0.947	-0.384 -0.883
psycho=~Q6_P	0.81	0.018	0.947	-0.545 -0.881
psycho=~Q7_P	0.94	0.004	0.947	-0.203 -0.897
psycho=~Q11_P	0.73	-0.006	0.947	-0.067 -0.892
psycho=~Q19_P	0.75	0.015	0.947	-0.138 -0.886
psycho=~Q26_P	0.63	0.026	0.947	-0.290 -0.886
physical=~Q3_F	0.84	0.022	0.895	-0.102 -0.886
physical=~Q4_F	0.83	0.010	0.895	-0.115 -0.913
physical=~Q10_F	0.63	0.010	0.895	0.012 -0.890
physical=~Q15_F	0.71	0.014	0.947	-0.030 -0.880
physical=~Q16_F	0.68	0.012	1.000	0.017 -0.873
physical=~Q17_F	0.89	0.001	1.000	-0.266 -0.894
physical=~Q18_F	0.61	-0.002	1.000	-0.131 -0.898
social=~Q20_S	0.94	-0.012	0.947	-0.015 -0.886
social=~Q21_S	0.89	-0.005	1.000	0.032 -0.885
social=~Q22_S	0.78	-0.007	0.895	-0.318 -0.877
environment=~Q8_A	0.37	0.013	0.895	0.031 -0.902

environment=~Q9_A	0.49	0.007	1.000	0.329	-0.930
environment=~Q12_A	0.79	0.007	1.000	0.225	-0.894
environment=~Q13_A	0.81	0.009	1.000	-0.054	-0.886
environment=~Q14_A	0.77	0.003	0.947	-0.294	-0.895
environment=~Q23_A	0.75	0.018	0.947	0.151	-0.890
environment=~Q24_A	0.64	-0.002	0.895	0.206	-0.902
environment=~Q25_A	0.73	-0.013	0.895	0.406	-0.911
Q3_F~~Q4_F	0.30	-0.004	1.000	0.142	-0.896
psycho~~physical	0.30	0.006	0.947	-0.312	-0.879
psycho~~social	0.30	-0.013	1.000	-0.008	-0.859
psycho~~environment	0.30	0.010	0.842	-0.201	-0.873
physical~~social	0.30	-0.014	1.000	-0.102	-0.875
physical~~environment	0.30	0.014	0.842	-0.419	-0.870
social~~environment	0.30	0.007	1.000	-0.311	-0.894
Q5_P~~Q5_P	0.15	-0.008	0.947	0.260	-0.858
Q6_P~~Q6_P	0.34	-0.003	1.000	0.225	-0.860
Q7_P~~Q7_P	0.12	0.007	0.895	-0.120	-0.872
Q11_P~~Q11_P	0.47	0.001	1.000	-0.279	-0.890
Q19_P~~Q19_P	0.44	0.001	0.895	-0.356	-0.847
Q26_P~~Q26_P	0.60	0.003	0.947	-0.116	-0.859
Q10_F~~Q10_F	0.60	-0.005	0.947	-0.024	-0.856
Q15_F~~Q15_F	0.50	0.003	1.000	-0.384	-0.867
Q16_F~~Q16_F	0.54	0.016	0.947	-0.270	-0.859
Q17_F~~Q17_F	0.21	0.001	1.000	0.110	-0.880
Q18_F~~Q18_F	0.63	0.007	0.947	0.231	-0.887
Q20_S~~Q20_S	0.12	-0.003	0.947	-0.479	-0.823
Q21_S~~Q21_S	0.21	-0.011	0.895	0.206	-0.817
Q22_S~~Q22_S	0.39	-0.003	0.947	0.175	-0.811
Q8_A~~Q8_A	0.86	-0.015	0.947	0.508	-0.896
Q9_A~~Q9_A	0.76	-0.021	0.947	0.441	-0.934
Q12_A~~Q12_A	0.38	-0.005	0.895	-0.016	-0.872
Q13_A~~Q13_A	0.34	-0.012	0.789	-0.380	-0.878
Q14_A~~Q14_A	0.41	-0.006	1.000	0.486	-0.903
Q23_A~~Q23_A	0.44	-0.006	1.000	-0.183	-0.879
Q24_A~~Q24_A	0.59	0.016	1.000	-0.431	-0.880
Q25_A~~Q25_A	0.47	-0.004	1.000	-0.179	-0.898
Q3_F~~Q3_F	1.00	-0.016	0.947	0.166	-0.875
Q4_F~~Q4_F	1.00	0.006	0.947	0.380	-0.898

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.355	-0.355	0.861	-0.708	-0.823	0.555	-0.354
aic	-0.355	1.000	1.000	-0.594	0.557	0.398	-0.867	1.000
bic	-0.355	1.000	1.000	-0.594	0.557	0.399	-0.867	1.000

```

rmsea  0.861 -0.594 -0.594  1.000 -0.937 -0.937  0.829 -0.594
cfi    -0.708  0.557  0.557 -0.937  1.000  0.955 -0.847  0.557
tli    -0.823  0.398  0.399 -0.937  0.955  1.000 -0.740  0.398
srmr   0.555 -0.867 -0.867  0.829 -0.847 -0.740  1.000 -0.866
n      -0.354  1.000  1.000 -0.594  0.557  0.398 -0.866  1.000
===== Replications =====
Number of replications = 19
Number of converged replications = 19
Number of nonconverged replications:
  1. Nonconvergent Results = 0
  2. Nonconvergent results from multiple imputation = 0
  3. At least one SE were negative or NA = 0
  4. Nonpositive-definite latent or observed (residual) covariance matrix
    (e.g., Heywood case or linear dependency) = 0
NOTE: The sample size is varying.

```

Source: [Simulation — Realistic Scenarios](#)

5.2 Missing Data (MCAR)

Missing data effectively shrinks N through a loss of information, meaning empirical sample sizes must often be adjusted proportionately. This simulation evaluates the explicit power penalty of 10% Missing Completely at Random strings.

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	312.298	5301.554	6038.158	0.052	0.938	0.930	0.074
2	325	299.574	17024.929	17563.472	0.041	0.954	0.948	0.061
3	550	286.849	28748.304	29088.787	0.029	0.969	0.965	0.048
4	775	274.125	40471.678	40614.102	0.018	0.984	0.983	0.035
5	1000	261.400	52195.053	52139.417	0.007	1.000	1.000	0.023

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5_P	0.927		0.045	0.039
psycho=~Q6_P	0.826		0.047	0.043
psycho=~Q7_P	0.944		0.042	0.039
psycho=~Q11_P	0.728		0.038	0.045

psycho=~Q19_P	0.766	0.050	0.045
psycho=~Q26_P	0.657	0.047	0.048
physical=~Q3_F	0.870	0.078	0.063
physical=~Q4_F	0.843	0.082	0.063
physical=~Q10_F	0.646	0.055	0.048
physical=~Q15_F	0.723	0.060	0.048
physical=~Q16_F	0.695	0.040	0.048
physical=~Q17_F	0.893	0.035	0.042
physical=~Q18_F	0.606	0.043	0.050
social=~Q20_S	0.928	0.047	0.040
social=~Q21_S	0.887	0.046	0.041
social=~Q22_S	0.771	0.038	0.044
environment=~Q8_A	0.383	0.050	0.052
environment=~Q9_A	0.496	0.042	0.050
environment=~Q12_A	0.796	0.040	0.045
environment=~Q13_A	0.816	0.033	0.043
environment=~Q14_A	0.769	0.051	0.045
environment=~Q23_A	0.770	0.042	0.046
environment=~Q24_A	0.637	0.071	0.049
environment=~Q25_A	0.720	0.063	0.045
Q3_F~~Q4_F	0.296	0.068	0.064
psycho~~physical	0.310	0.050	0.049
psycho~~social	0.288	0.057	0.049
psycho~~environment	0.311	0.061	0.049
physical~~social	0.283	0.056	0.051
physical~~environment	0.313	0.065	0.051
social~~environment	0.303	0.038	0.050
Q5_P~~Q5_P	0.142	0.020	0.017
Q6_P~~Q6_P	0.339	0.022	0.028
Q7_P~~Q7_P	0.127	0.024	0.017
Q11_P~~Q11_P	0.471	0.033	0.037
Q19_P~~Q19_P	0.443	0.050	0.036
Q26_P~~Q26_P	0.599	0.046	0.046
Q10_F~~Q10_F	0.581	0.045	0.047
Q15_F~~Q15_F	0.504	0.029	0.044
Q16_F~~Q16_F	0.550	0.043	0.046
Q17_F~~Q17_F	0.213	0.032	0.031
Q18_F~~Q18_F	0.641	0.043	0.051
Q20_S~~Q20_S	0.118	0.031	0.026
Q21_S~~Q21_S	0.197	0.030	0.027
Q22_S~~Q22_S	0.382	0.037	0.033
Q8_A~~Q8_A	0.849	0.061	0.063
Q9_A~~Q9_A	0.741	0.062	0.056

Q12_A~~Q12_A	0.379	0.036	0.035
Q13_A~~Q13_A	0.326	0.049	0.032
Q14_A~~Q14_A	0.402	0.041	0.036
Q23_A~~Q23_A	0.434	0.043	0.038
Q24_A~~Q24_A	0.608	0.045	0.049
Q25_A~~Q25_A	0.462	0.034	0.039
Q3_F~~Q3_F	0.983	0.082	0.082
Q4_F~~Q4_F	1.001	0.083	0.081
Q5_P~1	-0.012	0.060	0.050
Q6_P~1	-0.023	0.078	0.051
Q7_P~1	-0.011	0.062	0.050
Q11_P~1	-0.008	0.063	0.051
Q19_P~1	-0.009	0.044	0.051
Q26_P~1	-0.015	0.051	0.052
Q3_F~1	-0.033	0.071	0.067
Q4_F~1	-0.044	0.072	0.066
Q10_F~1	0.009	0.058	0.051
Q15_F~1	-0.027	0.052	0.052
Q16_F~1	-0.016	0.047	0.052
Q17_F~1	-0.032	0.049	0.051
Q18_F~1	-0.017	0.043	0.051
Q20_S~1	0.009	0.049	0.050
Q21_S~1	-0.008	0.073	0.049
Q22_S~1	0.000	0.054	0.050
Q8_A~1	0.004	0.056	0.051
Q9_A~1	0.005	0.060	0.050
Q12_A~1	0.011	0.049	0.050
Q13_A~1	0.002	0.037	0.050
Q14_A~1	-0.003	0.059	0.050
Q23_A~1	0.012	0.071	0.051
Q24_A~1	0.026	0.052	0.051
Q25_A~1	0.013	0.049	0.050

	Power (Not equal 0)	Std Est	Std Est	SD	Std Ave	SE
psycho=~Q5_P	1.000	0.926		0.011		0.010
psycho=~Q6_P	1.000	0.817		0.018		0.019
psycho=~Q7_P	1.000	0.935		0.014		0.010
psycho=~Q11_P	1.000	0.727		0.022		0.026
psycho=~Q19_P	1.000	0.755		0.020		0.024
psycho=~Q26_P	1.000	0.646		0.031		0.031
physical=~Q3_F	1.000	0.658		0.038		0.033
physical=~Q4_F	1.000	0.643		0.038		0.034
physical=~Q10_F	1.000	0.645		0.037		0.034
physical=~Q15_F	1.000	0.712		0.031		0.030

physical=~Q16_F	1.000	0.683	0.028	0.032
physical=~Q17_F	1.000	0.888	0.019	0.018
physical=~Q18_F	1.000	0.603	0.034	0.037
social=~Q20_S	1.000	0.937	0.019	0.015
social=~Q21_S	1.000	0.894	0.018	0.016
social=~Q22_S	1.000	0.780	0.027	0.023
environment=~Q8_A	1.000	0.383	0.042	0.047
environment=~Q9_A	1.000	0.499	0.029	0.042
environment=~Q12_A	1.000	0.791	0.022	0.024
environment=~Q13_A	1.000	0.819	0.024	0.021
environment=~Q14_A	1.000	0.771	0.028	0.024
environment=~Q23_A	1.000	0.760	0.021	0.026
environment=~Q24_A	1.000	0.630	0.050	0.035
environment=~Q25_A	1.000	0.725	0.041	0.029
Q3_F~~Q4_F	0.947	0.298	0.060	0.054
psycho~~physical	1.000	0.310	0.050	0.049
psycho~~social	0.947	0.288	0.057	0.049
psycho~~environment	1.000	0.311	0.061	0.049
physical~~social	1.000	0.283	0.056	0.051
physical~~environment	1.000	0.313	0.065	0.051
social~~environment	1.000	0.303	0.038	0.050
Q5_P~~Q5_P	1.000	0.142	0.021	0.019
Q6_P~~Q6_P	1.000	0.333	0.029	0.030
Q7_P~~Q7_P	1.000	0.125	0.026	0.018
Q11_P~~Q11_P	1.000	0.471	0.032	0.037
Q19_P~~Q19_P	1.000	0.429	0.031	0.036
Q26_P~~Q26_P	1.000	0.581	0.040	0.041
Q10_F~~Q10_F	1.000	0.582	0.047	0.044
Q15_F~~Q15_F	1.000	0.492	0.044	0.042
Q16_F~~Q16_F	1.000	0.532	0.039	0.043
Q17_F~~Q17_F	1.000	0.211	0.034	0.032
Q18_F~~Q18_F	1.000	0.635	0.041	0.044
Q20_S~~Q20_S	1.000	0.122	0.035	0.028
Q21_S~~Q21_S	1.000	0.201	0.033	0.029
Q22_S~~Q22_S	1.000	0.391	0.042	0.036
Q8_A~~Q8_A	1.000	0.852	0.034	0.036
Q9_A~~Q9_A	1.000	0.751	0.029	0.042
Q12_A~~Q12_A	1.000	0.374	0.034	0.037
Q13_A~~Q13_A	1.000	0.328	0.039	0.035
Q14_A~~Q14_A	1.000	0.405	0.044	0.038
Q23_A~~Q23_A	1.000	0.423	0.031	0.039
Q24_A~~Q24_A	1.000	0.600	0.059	0.043
Q25_A~~Q25_A	1.000	0.473	0.056	0.041

Q3_F~~Q3_F	1.000	0.565	0.050	0.044
Q4_F~~Q4_F	1.000	0.585	0.049	0.044
Q5_P~1	0.000	-0.011	0.059	0.050
Q6_P~1	0.053	-0.022	0.074	0.050
Q7_P~1	0.053	-0.011	0.061	0.050
Q11_P~1	0.000	-0.008	0.063	0.051
Q19_P~1	0.000	-0.008	0.043	0.050
Q26_P~1	0.000	-0.015	0.049	0.051
Q3_F~1	0.105	-0.026	0.053	0.051
Q4_F~1	0.053	-0.033	0.055	0.051
Q10_F~1	0.053	0.008	0.058	0.051
Q15_F~1	0.053	-0.026	0.050	0.051
Q16_F~1	0.000	-0.016	0.046	0.051
Q17_F~1	0.053	-0.032	0.049	0.050
Q18_F~1	0.000	-0.017	0.043	0.051
Q20_S~1	0.053	0.008	0.049	0.050
Q21_S~1	0.158	-0.009	0.075	0.050
Q22_S~1	0.053	-0.001	0.055	0.050
Q8_A~1	0.105	0.004	0.056	0.051
Q9_A~1	0.105	0.003	0.064	0.052
Q12_A~1	0.053	0.011	0.050	0.050
Q13_A~1	0.000	0.002	0.037	0.050
Q14_A~1	0.053	-0.004	0.060	0.051
Q23_A~1	0.105	0.012	0.071	0.051
Q24_A~1	0.053	0.024	0.051	0.051
Q25_A~1	0.000	0.012	0.051	0.051

	Average FMI1	SD FMI1	r_coef.n	r_se.n
psycho=~Q5_P	0.048	0.011	-0.308	-0.878
psycho=~Q6_P	0.067	0.014	-0.544	-0.887
psycho=~Q7_P	0.045	0.010	-0.196	-0.907
psycho=~Q11_P	0.076	0.012	-0.145	-0.899
psycho=~Q19_P	0.081	0.018	-0.307	-0.892
psycho=~Q26_P	0.088	0.023	-0.263	-0.873
physical=~Q3_F	0.084	0.023	-0.071	-0.868
physical=~Q4_F	0.092	0.027	-0.051	-0.902
physical=~Q10_F	0.104	0.025	-0.137	-0.886
physical=~Q15_F	0.086	0.028	-0.007	-0.874
physical=~Q16_F	0.096	0.021	0.141	-0.878
physical=~Q17_F	0.082	0.020	-0.404	-0.881
physical=~Q18_F	0.105	0.020	-0.104	-0.899
social=~Q20_S	0.075	0.022	0.004	-0.886
social=~Q21_S	0.076	0.032	-0.059	-0.878
social=~Q22_S	0.081	0.026	-0.167	-0.871

environment=~Q8_A	0.112	0.031	0.048	-0.899
environment=~Q9_A	0.105	0.023	0.412	-0.928
environment=~Q12_A	0.083	0.022	0.394	-0.885
environment=~Q13_A	0.076	0.015	0.069	-0.893
environment=~Q14_A	0.084	0.021	-0.207	-0.895
environment=~Q23_A	0.084	0.017	-0.009	-0.894
environment=~Q24_A	0.099	0.027	0.231	-0.894
environment=~Q25_A	0.085	0.022	0.345	-0.903
Q3_F~~Q4_F	0.183	0.032	-0.024	-0.878
psycho~~physical	0.028	0.006	-0.395	-0.877
psycho~~social	0.028	0.011	-0.041	-0.858
psycho~~environment	0.024	0.007	-0.221	-0.873
physical~~social	0.038	0.013	-0.137	-0.875
physical~~environment	0.032	0.009	-0.413	-0.869
social~~environment	0.034	0.015	-0.230	-0.886
Q5_P~~Q5_P	0.189	0.020	0.247	-0.872
Q6_P~~Q6_P	0.136	0.021	0.285	-0.880
Q7_P~~Q7_P	0.195	0.029	-0.063	-0.878
Q11_P~~Q11_P	0.120	0.013	-0.284	-0.888
Q19_P~~Q19_P	0.125	0.027	-0.306	-0.860
Q26_P~~Q26_P	0.111	0.018	-0.087	-0.849
Q10_F~~Q10_F	0.126	0.022	0.188	-0.864
Q15_F~~Q15_F	0.141	0.025	-0.506	-0.855
Q16_F~~Q16_F	0.134	0.024	-0.142	-0.873
Q17_F~~Q17_F	0.185	0.018	0.264	-0.858
Q18_F~~Q18_F	0.121	0.013	0.134	-0.887
Q20_S~~Q20_S	0.222	0.025	-0.437	-0.826
Q21_S~~Q21_S	0.203	0.038	0.423	-0.796
Q22_S~~Q22_S	0.149	0.023	0.205	-0.798
Q8_A~~Q8_A	0.102	0.024	0.402	-0.890
Q9_A~~Q9_A	0.117	0.022	0.446	-0.939
Q12_A~~Q12_A	0.135	0.011	-0.262	-0.864
Q13_A~~Q13_A	0.143	0.021	-0.329	-0.884
Q14_A~~Q14_A	0.146	0.032	0.479	-0.894
Q23_A~~Q23_A	0.140	0.023	-0.160	-0.878
Q24_A~~Q24_A	0.123	0.023	-0.396	-0.889
Q25_A~~Q25_A	0.127	0.021	-0.196	-0.883
Q3_F~~Q3_F	0.133	0.047	-0.022	-0.855
Q4_F~~Q4_F	0.125	0.022	0.287	-0.885
Q5_P~1	0.023	0.005	0.157	-0.875
Q6_P~1	0.041	0.008	0.294	-0.885
Q7_P~1	0.023	0.004	0.221	-0.910
Q11_P~1	0.052	0.009	0.351	-0.895

Q19_P~1	0.052	0.011	0.145	-0.895
Q26_P~1	0.064	0.010	-0.132	-0.877
Q3_F~1	0.057	0.008	-0.065	-0.870
Q4_F~1	0.060	0.012	0.170	-0.906
Q10_F~1	0.068	0.009	-0.212	-0.882
Q15_F~1	0.060	0.012	0.382	-0.867
Q16_F~1	0.064	0.008	0.140	-0.873
Q17_F~1	0.040	0.006	0.503	-0.893
Q18_F~1	0.071	0.010	0.310	-0.899
Q20_S~1	0.031	0.009	-0.018	-0.893
Q21_S~1	0.032	0.008	0.263	-0.892
Q22_S~1	0.046	0.007	0.206	-0.894
Q8_A~1	0.083	0.014	-0.033	-0.897
Q9_A~1	0.085	0.017	0.460	-0.924
Q12_A~1	0.045	0.008	0.053	-0.890
Q13_A~1	0.042	0.009	0.079	-0.889
Q14_A~1	0.052	0.009	0.004	-0.888
Q23_A~1	0.054	0.009	0.136	-0.884
Q24_A~1	0.069	0.013	-0.099	-0.909
Q25_A~1	0.056	0.012	0.008	-0.910

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.389	-0.390	0.943	-0.812	-0.859	0.643	-0.387
aic	-0.389	1.000	1.000	-0.531	0.452	0.401	-0.870	1.000
bic	-0.390	1.000	1.000	-0.532	0.453	0.401	-0.870	1.000
rmsea	0.943	-0.531	-0.532	1.000	-0.936	-0.954	0.803	-0.530
cfi	-0.812	0.452	0.453	-0.936	1.000	0.991	-0.771	0.452
tli	-0.859	0.401	0.401	-0.954	0.991	1.000	-0.737	0.400
srmr	0.643	-0.870	-0.870	0.803	-0.771	-0.737	1.000	-0.869
n	-0.387	1.000	1.000	-0.530	0.452	0.400	-0.869	1.000

===== Replications =====

Number of replications = 19

Number of converged replications = 19

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

NOTE: The data generation model is not the same as the analysis model. See the summary of the

Source: [Simulation — Realistic Scenarios](#)

5.3 Sensitivity Across Population Models

A robust power analysis examines boundaries. How drastically does necessary N shift when our assumption about structural quality is overly naive, or overly optimistic?

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	304.405	6683.676	6849.187	0.033	0.927	0.918	0.079
2	325	288.104	21383.339	21584.617	0.025	0.946	0.939	0.066
3	550	271.802	36083.001	36320.047	0.017	0.964	0.960	0.052
4	775	255.501	50782.664	51055.477	0.008	0.982	0.981	0.039
5	1000	239.200	65482.326	65790.907	0.000	1.000	1.002	0.025

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5_P	0.497		0.053	0.055
psycho=~Q6_P	0.524		0.038	0.056
psycho=~Q7_P	0.508		0.057	0.056
psycho=~Q11_P	0.494		0.070	0.055
psycho=~Q19_P	0.501		0.063	0.056
psycho=~Q26_P	0.494		0.072	0.055
physical=~Q3_F	0.506		0.060	0.056
physical=~Q4_F	0.502		0.062	0.056
physical=~Q10_F	0.500		0.065	0.055
physical=~Q15_F	0.498		0.054	0.055
physical=~Q16_F	0.508		0.064	0.056
physical=~Q17_F	0.489		0.047	0.054
physical=~Q18_F	0.538		0.077	0.056
social=~Q20_S	0.517		0.067	0.067
social=~Q21_S	0.521		0.081	0.067
social=~Q22_S	0.519		0.064	0.067
environment=~Q8_A	0.505		0.079	0.054
environment=~Q9_A	0.487		0.066	0.054
environment=~Q12_A	0.506		0.053	0.053
environment=~Q13_A	0.485		0.047	0.054
environment=~Q14_A	0.525		0.057	0.054
environment=~Q23_A	0.507		0.055	0.054
environment=~Q24_A	0.522		0.053	0.054
environment=~Q25_A	0.488		0.069	0.053

Q3_F~~Q4_F	0.300	0.041	0.049
psycho~~physical	0.287	0.062	0.067
psycho~~social	0.334	0.100	0.076
psycho~~environment	0.305	0.077	0.065
physical~~social	0.317	0.074	0.076
physical~~environment	0.307	0.098	0.064
social~~environment	0.312	0.070	0.075
Q5_P~~Q5_P	0.732	0.050	0.060
Q6_P~~Q6_P	0.728	0.062	0.061
Q7_P~~Q7_P	0.740	0.060	0.061
Q11_P~~Q11_P	0.744	0.075	0.060
Q19_P~~Q19_P	0.743	0.048	0.061
Q26_P~~Q26_P	0.720	0.058	0.058
Q10_F~~Q10_F	0.742	0.061	0.060
Q15_F~~Q15_F	0.746	0.053	0.060
Q16_F~~Q16_F	0.750	0.079	0.062
Q17_F~~Q17_F	0.742	0.057	0.059
Q18_F~~Q18_F	0.741	0.069	0.062
Q20_S~~Q20_S	0.725	0.061	0.072
Q21_S~~Q21_S	0.746	0.077	0.073
Q22_S~~Q22_S	0.720	0.074	0.071
Q8_A~~Q8_A	0.756	0.076	0.060
Q9_A~~Q9_A	0.771	0.080	0.061
Q12_A~~Q12_A	0.732	0.053	0.058
Q13_A~~Q13_A	0.767	0.067	0.060
Q14_A~~Q14_A	0.748	0.065	0.060
Q23_A~~Q23_A	0.756	0.061	0.060
Q24_A~~Q24_A	0.740	0.057	0.060
Q25_A~~Q25_A	0.748	0.064	0.059
Q3_F~~Q3_F	0.752	0.049	0.062
Q4_F~~Q4_F	0.756	0.052	0.062

	Power (Not equal 0)	Std Est	Std Est SD	Std Ave SE
psycho=~Q5_P	1.000	0.501	0.041	0.049
psycho=~Q6_P	1.000	0.523	0.033	0.048
psycho=~Q7_P	1.000	0.507	0.045	0.049
psycho=~Q11_P	1.000	0.496	0.063	0.049
psycho=~Q19_P	1.000	0.501	0.055	0.049
psycho=~Q26_P	1.000	0.500	0.061	0.048
physical=~Q3_F	1.000	0.503	0.052	0.048
physical=~Q4_F	1.000	0.499	0.055	0.049
physical=~Q10_F	1.000	0.501	0.058	0.048
physical=~Q15_F	1.000	0.498	0.041	0.048
physical=~Q16_F	1.000	0.505	0.060	0.048

physical=~Q17_F	1.000	0.493	0.038	0.048
physical=~Q18_F	1.000	0.528	0.064	0.046
social=~Q20_S	1.000	0.517	0.055	0.060
social=~Q21_S	1.000	0.514	0.075	0.060
social=~Q22_S	1.000	0.521	0.055	0.061
environment=~Q8_A	1.000	0.500	0.077	0.046
environment=~Q9_A	1.000	0.484	0.061	0.047
environment=~Q12_A	1.000	0.508	0.041	0.045
environment=~Q13_A	1.000	0.485	0.044	0.047
environment=~Q14_A	1.000	0.518	0.045	0.045
environment=~Q23_A	1.000	0.502	0.038	0.046
environment=~Q24_A	1.000	0.518	0.041	0.045
environment=~Q25_A	1.000	0.490	0.062	0.046
Q3_F~~Q4_F	1.000	0.397	0.042	0.047
psycho~~physical	0.947	0.287	0.062	0.067
psycho~~social	0.947	0.334	0.100	0.076
psycho~~environment	0.947	0.305	0.077	0.065
physical~~social	0.895	0.317	0.074	0.076
physical~~environment	0.947	0.307	0.098	0.064
social~~environment	0.947	0.312	0.070	0.075
Q5_P~~Q5_P	1.000	0.747	0.042	0.049
Q6_P~~Q6_P	1.000	0.725	0.035	0.050
Q7_P~~Q7_P	1.000	0.741	0.047	0.049
Q11_P~~Q11_P	1.000	0.750	0.066	0.048
Q19_P~~Q19_P	1.000	0.746	0.055	0.048
Q26_P~~Q26_P	1.000	0.746	0.064	0.049
Q10_F~~Q10_F	1.000	0.746	0.058	0.047
Q15_F~~Q15_F	1.000	0.750	0.042	0.047
Q16_F~~Q16_F	1.000	0.741	0.057	0.047
Q17_F~~Q17_F	1.000	0.756	0.036	0.047
Q18_F~~Q18_F	1.000	0.717	0.073	0.050
Q20_S~~Q20_S	1.000	0.730	0.057	0.063
Q21_S~~Q21_S	1.000	0.730	0.076	0.062
Q22_S~~Q22_S	1.000	0.726	0.058	0.064
Q8_A~~Q8_A	1.000	0.744	0.066	0.044
Q9_A~~Q9_A	1.000	0.762	0.056	0.044
Q12_A~~Q12_A	1.000	0.740	0.042	0.046
Q13_A~~Q13_A	1.000	0.763	0.043	0.045
Q14_A~~Q14_A	1.000	0.730	0.048	0.047
Q23_A~~Q23_A	1.000	0.746	0.039	0.046
Q24_A~~Q24_A	1.000	0.730	0.042	0.047
Q25_A~~Q25_A	1.000	0.756	0.056	0.044
Q3_F~~Q3_F	1.000	0.744	0.053	0.049

Q4_F~~Q4_F	1.000	0.748	0.057	0.048
	Average Param	Average Bias	Coverage	r_coef.n r_se.n
psycho=~Q5_P	0.50	-0.003	0.947	-0.266 -0.902
psycho=~Q6_P	0.50	0.024	1.000	-0.083 -0.877
psycho=~Q7_P	0.50	0.008	0.947	0.352 -0.904
psycho=~Q11_P	0.50	-0.006	0.842	-0.223 -0.906
psycho=~Q19_P	0.50	0.001	1.000	-0.073 -0.895
psycho=~Q26_P	0.50	-0.006	0.895	-0.122 -0.889
physical=~Q3_F	0.50	0.006	1.000	-0.118 -0.896
physical=~Q4_F	0.50	0.002	0.895	-0.014 -0.882
physical=~Q10_F	0.50	0.000	1.000	0.033 -0.899
physical=~Q15_F	0.50	-0.002	1.000	0.210 -0.922
physical=~Q16_F	0.50	0.008	0.947	0.118 -0.879
physical=~Q17_F	0.50	-0.011	0.947	0.047 -0.945
physical=~Q18_F	0.50	0.038	0.842	-0.392 -0.881
social=~Q20_S	0.50	0.017	1.000	-0.363 -0.881
social=~Q21_S	0.50	0.021	0.895	-0.152 -0.928
social=~Q22_S	0.50	0.019	0.947	-0.063 -0.925
environment=~Q8_A	0.50	0.005	0.895	0.158 -0.883
environment=~Q9_A	0.50	-0.013	1.000	0.394 -0.851
environment=~Q12_A	0.50	0.006	1.000	-0.181 -0.878
environment=~Q13_A	0.50	-0.015	0.947	0.236 -0.901
environment=~Q14_A	0.50	0.025	0.947	-0.286 -0.879
environment=~Q23_A	0.50	0.007	0.947	0.110 -0.875
environment=~Q24_A	0.50	0.022	0.947	-0.322 -0.865
environment=~Q25_A	0.50	-0.012	0.947	0.239 -0.905
Q3_F~~Q4_F	0.30	0.000	0.947	0.008 -0.871
psycho~~physical	0.30	-0.013	0.947	-0.056 -0.899
psycho~~social	0.30	0.034	0.842	-0.232 -0.915
psycho~~environment	0.30	0.005	0.842	-0.476 -0.865
physical~~social	0.30	0.017	0.947	-0.038 -0.896
physical~~environment	0.30	0.007	0.895	-0.318 -0.864
social~~environment	0.30	0.012	0.895	-0.110 -0.895
Q5_P~~Q5_P	0.75	-0.018	0.947	-0.246 -0.899
Q6_P~~Q6_P	0.75	-0.022	0.947	-0.384 -0.846
Q7_P~~Q7_P	0.75	-0.010	0.947	-0.025 -0.893
Q11_P~~Q11_P	0.75	-0.006	0.895	0.282 -0.906
Q19_P~~Q19_P	0.75	-0.007	1.000	-0.175 -0.890
Q26_P~~Q26_P	0.75	-0.030	1.000	0.518 -0.881
Q10_F~~Q10_F	0.75	-0.008	1.000	0.327 -0.881
Q15_F~~Q15_F	0.75	-0.004	1.000	-0.072 -0.926
Q16_F~~Q16_F	0.75	0.000	0.895	-0.254 -0.841
Q17_F~~Q17_F	0.75	-0.008	1.000	0.093 -0.967

Q18_F~~Q18_F	0.75	-0.009	0.947	0.460	-0.849
Q20_S~~Q20_S	0.75	-0.025	0.947	0.197	-0.831
Q21_S~~Q21_S	0.75	-0.004	0.947	0.261	-0.921
Q22_S~~Q22_S	0.75	-0.030	0.895	0.327	-0.910
Q8_A~~Q8_A	0.75	0.006	0.947	-0.111	-0.865
Q9_A~~Q9_A	0.75	0.021	0.947	-0.295	-0.805
Q12_A~~Q12_A	0.75	-0.018	0.947	0.319	-0.859
Q13_A~~Q13_A	0.75	0.017	1.000	-0.188	-0.896
Q14_A~~Q14_A	0.75	-0.002	0.947	0.037	-0.864
Q23_A~~Q23_A	0.75	0.006	1.000	-0.308	-0.853
Q24_A~~Q24_A	0.75	-0.010	0.895	-0.081	-0.839
Q25_A~~Q25_A	0.75	-0.002	0.947	-0.230	-0.908
Q3_F~~Q3_F	0.75	0.002	1.000	0.034	-0.887
Q4_F~~Q4_F	0.75	0.006	1.000	0.092	-0.862

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.302	-0.302	0.885	-0.742	-0.858	0.443	-0.301
aic	-0.302	1.000	1.000	-0.575	0.602	0.371	-0.882	1.000
bic	-0.302	1.000	1.000	-0.575	0.602	0.371	-0.883	1.000
rmsea	0.885	-0.575	-0.575	1.000	-0.956	-0.943	0.771	-0.576
cfi	-0.742	0.602	0.602	-0.956	1.000	0.939	-0.853	0.603
tli	-0.858	0.371	0.371	-0.943	0.939	1.000	-0.659	0.371
srmr	0.443	-0.882	-0.883	0.771	-0.853	-0.659	1.000	-0.883
n	-0.301	1.000	1.000	-0.576	0.603	0.371	-0.883	1.000

===== Replications =====

Number of replications = 19

Number of converged replications = 19

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Realistic Scenarios](#)

RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

	N	chisq	aic	bic	rmsea	cfi	tli	srmr
1	100	312.967	5788.754	5865.234	0.040	0.969	0.966	0.066
2	325	296.819	18226.819	18383.582	0.030	0.977	0.974	0.054
3	550	280.672	30664.884	30901.929	0.020	0.985	0.983	0.043
4	775	264.524	43102.948	43420.276	0.011	0.992	0.992	0.031
5	1000	248.376	55541.013	55938.624	0.001	1.000	1.001	0.020

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5_P	0.703	0.033	0.045	
psycho=~Q6_P	0.701	0.040	0.045	
psycho=~Q7_P	0.709	0.058	0.045	
psycho=~Q11_P	0.710	0.036	0.046	
psycho=~Q19_P	0.686	0.043	0.045	
psycho=~Q26_P	0.705	0.048	0.046	
physical=~Q3_F	0.707	0.047	0.046	
physical=~Q4_F	0.706	0.048	0.046	
physical=~Q10_F	0.718	0.065	0.045	
physical=~Q15_F	0.706	0.047	0.045	
physical=~Q16_F	0.720	0.047	0.046	
physical=~Q17_F	0.683	0.062	0.045	
physical=~Q18_F	0.712	0.050	0.045	
social=~Q20_S	0.715	0.045	0.050	
social=~Q21_S	0.728	0.074	0.050	
social=~Q22_S	0.712	0.041	0.049	
environment=~Q8_A	0.695	0.055	0.045	
environment=~Q9_A	0.706	0.044	0.045	
environment=~Q12_A	0.717	0.037	0.046	
environment=~Q13_A	0.709	0.046	0.044	
environment=~Q14_A	0.702	0.063	0.044	
environment=~Q23_A	0.698	0.060	0.044	
environment=~Q24_A	0.699	0.035	0.045	
environment=~Q25_A	0.714	0.032	0.045	
Q3_F~~Q4_F	0.502	0.032	0.042	
psycho~~physical	0.499	0.043	0.045	
psycho~~social	0.525	0.056	0.047	
psycho~~environment	0.506	0.055	0.043	
physical~~social	0.512	0.053	0.048	
physical~~environment	0.504	0.063	0.043	
social~~environment	0.510	0.049	0.048	
Q5_P~~Q5_P	0.504	0.049	0.041	
Q6_P~~Q6_P	0.508	0.048	0.041	
Q7_P~~Q7_P	0.496	0.053	0.040	
Q11_P~~Q11_P	0.510	0.037	0.042	

Q19_P~~Q19_P	0.509	0.037	0.041
Q26_P~~Q26_P	0.511	0.034	0.042
Q10_F~~Q10_F	0.494	0.033	0.041
Q15_F~~Q15_F	0.498	0.038	0.041
Q16_F~~Q16_F	0.508	0.034	0.042
Q17_F~~Q17_F	0.520	0.045	0.042
Q18_F~~Q18_F	0.501	0.035	0.041
Q20_S~~Q20_S	0.524	0.054	0.051
Q21_S~~Q21_S	0.497	0.077	0.050
Q22_S~~Q22_S	0.502	0.045	0.049
Q8_A~~Q8_A	0.512	0.051	0.040
Q9_A~~Q9_A	0.512	0.033	0.040
Q12_A~~Q12_A	0.526	0.032	0.042
Q13_A~~Q13_A	0.491	0.028	0.039
Q14_A~~Q14_A	0.494	0.042	0.039
Q23_A~~Q23_A	0.490	0.043	0.039
Q24_A~~Q24_A	0.519	0.042	0.041
Q25_A~~Q25_A	0.510	0.032	0.040
Q3_F~~Q3_F	0.511	0.033	0.042
Q4_F~~Q4_F	0.512	0.032	0.042

	Power (Not equal 0)	Std Est	Std Est SD	Std Ave SE
psycho=~Q5_P	1	0.703	0.025	0.029
psycho=~Q6_P	1	0.701	0.029	0.029
psycho=~Q7_P	1	0.708	0.043	0.029
psycho=~Q11_P	1	0.705	0.028	0.029
psycho=~Q19_P	1	0.693	0.024	0.030
psycho=~Q26_P	1	0.701	0.030	0.029
physical=~Q3_F	1	0.702	0.031	0.029
physical=~Q4_F	1	0.701	0.031	0.029
physical=~Q10_F	1	0.713	0.033	0.029
physical=~Q15_F	1	0.706	0.032	0.029
physical=~Q16_F	1	0.710	0.027	0.029
physical=~Q17_F	1	0.686	0.034	0.031
physical=~Q18_F	1	0.708	0.029	0.029
social=~Q20_S	1	0.702	0.032	0.035
social=~Q21_S	1	0.717	0.047	0.034
social=~Q22_S	1	0.709	0.022	0.034
environment=~Q8_A	1	0.696	0.041	0.029
environment=~Q9_A	1	0.702	0.026	0.028
environment=~Q12_A	1	0.703	0.019	0.028
environment=~Q13_A	1	0.710	0.029	0.027
environment=~Q14_A	1	0.705	0.042	0.028
environment=~Q23_A	1	0.705	0.038	0.028

environment=~Q24_A	1	0.696	0.020	0.029
environment=~Q25_A	1	0.707	0.019	0.028
Q3_F~~Q4_F	1	0.980	0.002	0.002
psycho~~physical	1	0.499	0.043	0.045
psycho~~social	1	0.525	0.056	0.047
psycho~~environment	1	0.506	0.055	0.043
physical~~social	1	0.512	0.053	0.048
physical~~environment	1	0.504	0.063	0.043
social~~environment	1	0.510	0.049	0.048
Q5_P~~Q5_P	1	0.505	0.035	0.041
Q6_P~~Q6_P	1	0.508	0.041	0.041
Q7_P~~Q7_P	1	0.497	0.061	0.040
Q11_P~~Q11_P	1	0.503	0.040	0.041
Q19_P~~Q19_P	1	0.520	0.033	0.041
Q26_P~~Q26_P	1	0.508	0.042	0.041
Q10_F~~Q10_F	1	0.491	0.046	0.041
Q15_F~~Q15_F	1	0.500	0.045	0.041
Q16_F~~Q16_F	1	0.496	0.039	0.041
Q17_F~~Q17_F	1	0.528	0.045	0.041
Q18_F~~Q18_F	1	0.498	0.042	0.041
Q20_S~~Q20_S	1	0.506	0.045	0.048
Q21_S~~Q21_S	1	0.484	0.067	0.048
Q22_S~~Q22_S	1	0.497	0.031	0.048
Q8_A~~Q8_A	1	0.514	0.054	0.039
Q9_A~~Q9_A	1	0.507	0.037	0.040
Q12_A~~Q12_A	1	0.506	0.027	0.040
Q13_A~~Q13_A	1	0.495	0.041	0.039
Q14_A~~Q14_A	1	0.501	0.056	0.039
Q23_A~~Q23_A	1	0.502	0.053	0.039
Q24_A~~Q24_A	1	0.515	0.028	0.040
Q25_A~~Q25_A	1	0.500	0.027	0.039
Q3_F~~Q3_F	1	0.506	0.043	0.041
Q4_F~~Q4_F	1	0.507	0.043	0.041
Average Param Average Bias Coverage r_coef.n r_se.n				
psycho=~Q5_P	0.70	0.003	1.000	-0.309 -0.899
psycho=~Q6_P	0.70	0.001	0.947	-0.191 -0.905
psycho=~Q7_P	0.70	0.009	0.947	-0.226 -0.901
psycho=~Q11_P	0.70	0.010	0.947	-0.135 -0.895
psycho=~Q19_P	0.70	-0.014	1.000	0.075 -0.908
psycho=~Q26_P	0.70	0.005	1.000	-0.454 -0.894
physical=~Q3_F	0.70	0.007	0.947	-0.153 -0.889
physical=~Q4_F	0.70	0.006	0.947	-0.137 -0.887
physical=~Q10_F	0.70	0.018	0.842	0.061 -0.876

physical=~Q15_F	0.70	0.006	1.000	-0.204	-0.897
physical=~Q16_F	0.70	0.020	1.000	-0.302	-0.896
physical=~Q17_F	0.70	-0.017	0.842	0.144	-0.919
physical=~Q18_F	0.70	0.012	0.895	-0.179	-0.910
social=~Q20_S	0.70	0.015	0.947	0.036	-0.902
social=~Q21_S	0.70	0.028	0.947	-0.295	-0.896
social=~Q22_S	0.70	0.012	1.000	-0.237	-0.909
environment=~Q8_A	0.70	-0.005	1.000	0.153	-0.872
environment=~Q9_A	0.70	0.006	0.947	0.130	-0.880
environment=~Q12_A	0.70	0.017	1.000	-0.244	-0.877
environment=~Q13_A	0.70	0.009	1.000	-0.337	-0.889
environment=~Q14_A	0.70	0.002	0.947	0.238	-0.892
environment=~Q23_A	0.70	-0.002	0.895	0.181	-0.897
environment=~Q24_A	0.70	-0.001	1.000	-0.237	-0.860
environment=~Q25_A	0.70	0.014	1.000	-0.050	-0.886
Q3_F~~Q4_F	0.50	0.002	1.000	0.073	-0.881
psycho~~physical	0.50	-0.001	0.947	-0.209	-0.881
psycho~~social	0.50	0.025	0.895	-0.371	-0.915
psycho~~environment	0.50	0.006	0.895	-0.458	-0.836
physical~~social	0.50	0.012	0.947	-0.182	-0.878
physical~~environment	0.50	0.004	0.895	-0.383	-0.826
social~~environment	0.50	0.010	0.895	-0.218	-0.868
Q5_P~~Q5_P	0.51	-0.006	0.895	-0.086	-0.879
Q6_P~~Q6_P	0.51	-0.002	0.947	0.085	-0.883
Q7_P~~Q7_P	0.51	-0.014	0.895	0.384	-0.888
Q11_P~~Q11_P	0.51	0.000	1.000	0.107	-0.893
Q19_P~~Q19_P	0.51	-0.001	1.000	0.210	-0.917
Q26_P~~Q26_P	0.51	0.001	0.895	-0.121	-0.891
Q10_F~~Q10_F	0.51	-0.016	0.895	0.006	-0.862
Q15_F~~Q15_F	0.51	-0.012	0.895	-0.006	-0.875
Q16_F~~Q16_F	0.51	-0.002	1.000	0.297	-0.893
Q17_F~~Q17_F	0.51	0.010	0.947	-0.244	-0.911
Q18_F~~Q18_F	0.51	-0.009	1.000	0.415	-0.923
Q20_S~~Q20_S	0.51	0.014	0.947	-0.362	-0.894
Q21_S~~Q21_S	0.51	-0.013	0.895	-0.050	-0.866
Q22_S~~Q22_S	0.51	-0.008	1.000	0.062	-0.897
Q8_A~~Q8_A	0.51	0.002	0.895	-0.143	-0.805
Q9_A~~Q9_A	0.51	0.002	0.947	0.003	-0.849
Q12_A~~Q12_A	0.51	0.016	1.000	-0.240	-0.853
Q13_A~~Q13_A	0.51	-0.019	1.000	0.371	-0.869
Q14_A~~Q14_A	0.51	-0.016	0.842	-0.083	-0.852
Q23_A~~Q23_A	0.51	-0.020	0.895	-0.055	-0.868
Q24_A~~Q24_A	0.51	0.009	1.000	-0.399	-0.817

Q25_A~~Q25_A	0.51	0.000	0.947	-0.162	-0.869
Q3_F~~Q3_F	0.51	0.001	1.000	0.065	-0.884
Q4_F~~Q4_F	0.51	0.002	1.000	0.084	-0.878

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr	n
chisq	1.000	-0.294	-0.294	0.876	-0.728	-0.834	0.524	-0.294
aic	-0.294	1.000	1.000	-0.574	0.582	0.401	-0.868	1.000
bic	-0.294	1.000	1.000	-0.574	0.583	0.401	-0.868	1.000
rmsea	0.876	-0.574	-0.574	1.000	-0.945	-0.943	0.822	-0.575
cfi	-0.728	0.582	0.583	-0.945	1.000	0.959	-0.885	0.583
tli	-0.834	0.401	0.401	-0.943	0.959	1.000	-0.747	0.401
srmr	0.524	-0.868	-0.868	0.822	-0.885	-0.747	1.000	-0.868
n	-0.294	1.000	1.000	-0.575	0.583	0.401	-0.868	1.000

===== Replications =====

Number of replications = 19

Number of converged replications = 19

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

NOTE: The sample size is varying.

Source: [Simulation — Realistic Scenarios](#)

5.4 Comparison

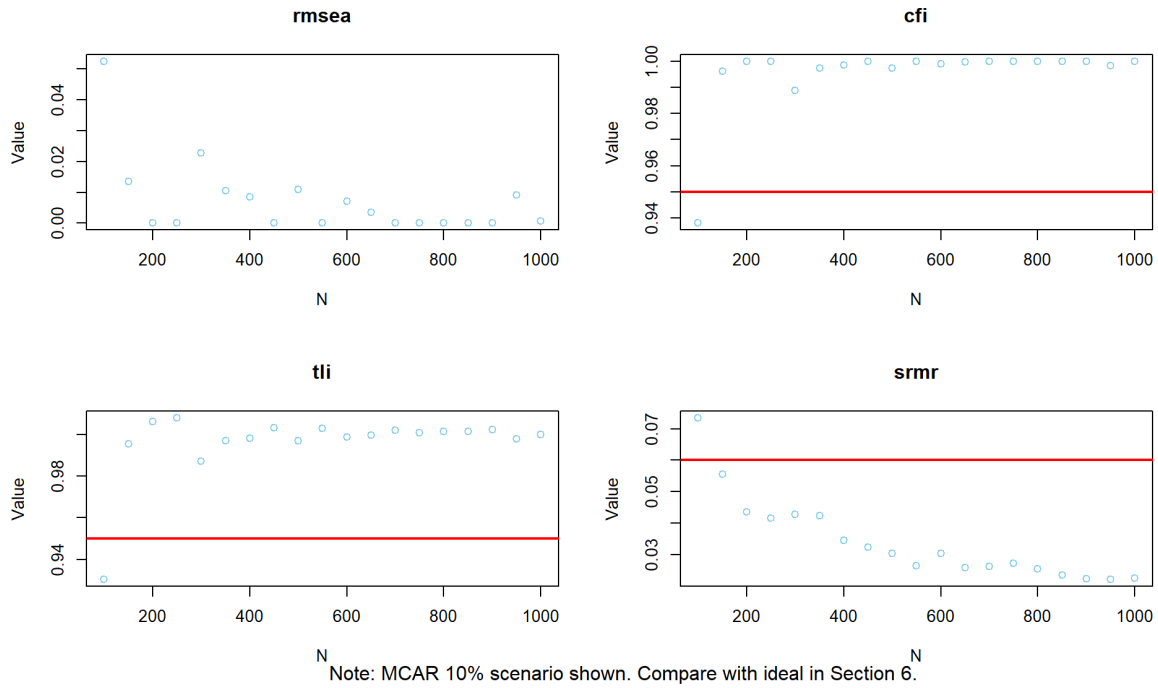


Figure 4: Power curves across population model scenarios (fit-index based)

Source: [Simulation — Realistic Scenarios](#)

Table 5: Required N for 80% power across all analysis scenarios

Scenario	Approach	N for 80% Power	Key Assumption
Analytical (semPower, RMSEA = .05)	Analytical	See Section 3.2	Continuous, normal, ML
Simulation — Ideal (popModel, ML)	Monte Carlo	From Section 6	Continuous, normal, ML
Simulation — Missing data (popModel, 10% MCAR)	Monte Carlo	From sim-missing above	Continuous, normal, ML + 10% MCAR
Simulation — Naive model (loadings = .5)	Monte Carlo	From sim-naive above	Weak loadings (0.5)
Simulation — Optimistic model (loadings = .7)	Monte Carlo	From sim-opt above	Strong loadings (0.7), mod. correlations (0.5)

6 Population Model vs. Empirical Model

Consider an epistemological question driving CFA: can we statistically distinguish the theorized population structure from an empirically estimated model with trivial incidental misspecifications that are ubiquitous in real-life sample covariance matrices? In this section, we compare empirical power distributions generated under a “Null” (empirical sample parameters representing noise) and an “Alternative” (our clean meta-analytic population model).

Null model simulation summary:RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

Alpha						
Fit Indices	0.1	0.05	0.01	0.001	Mean	SD
chisq	276.046	281.671	301.135	306.909	252.587	21.309
aic	59722.838	59761.746	59850.452	59883.548	59438.795	212.848
bic	59995.291	60034.199	60122.904	60156.001	59711.248	212.848
rmsea	0.011	0.012	0.015	0.016	0.005	0.005
cfi	0.996	0.996	0.993	0.992	0.999	0.002
tli	0.996	0.995	0.992	0.991	0.999	0.003
srmr	0.022	0.022	0.023	0.023	0.021	0.001

===== Parameter Estimates and Standard Errors =====

	Estimate	Average Estimate	SD	Average SE
psycho=~Q5_P	0.523	0.024	0.024	
psycho=~Q6_P	0.628	0.026	0.028	
psycho=~Q7_P	0.547	0.020	0.025	
psycho=~Q11_P	0.597	0.033	0.029	
psycho=~Q19_P	0.778	0.023	0.025	
psycho=~Q26_P	0.545	0.025	0.027	
physical=~Q3_F	0.415	0.037	0.029	
physical=~Q4_F	0.387	0.033	0.031	
physical=~Q10_F	0.670	0.026	0.025	
physical=~Q15_F	0.392	0.019	0.024	
physical=~Q16_F	0.553	0.030	0.031	
physical=~Q17_F	0.812	0.024	0.023	
physical=~Q18_F	0.812	0.023	0.024	
social=~Q20_S	0.702	0.022	0.027	
social=~Q21_S	0.501	0.035	0.033	
social=~Q22_S	0.633	0.024	0.027	

environment=~Q8_A	0.446	0.028	0.031			
environment=~Q9_A	0.512	0.025	0.028			
environment=~Q12_A	0.750	0.034	0.033			
environment=~Q13_A	0.497	0.023	0.025			
environment=~Q14_A	0.648	0.028	0.028			
environment=~Q23_A	0.493	0.026	0.030			
environment=~Q24_A	0.469	0.031	0.031			
environment=~Q25_A	0.466	0.034	0.031			
Q3_F~~Q4_F	0.339	0.029	0.027			
Q5_P~~Q5_P	0.401	0.017	0.019			
Q6_P~~Q6_P	0.582	0.025	0.027			
Q7_P~~Q7_P	0.439	0.018	0.021			
Q11_P~~Q11_P	0.627	0.024	0.029			
Q19_P~~Q19_P	0.292	0.016	0.017			
Q26_P~~Q26_P	0.558	0.026	0.026			
Q3_F~~Q3_F	0.703	0.035	0.031			
Q4_F~~Q4_F	0.861	0.036	0.038			
Q10_F~~Q10_F	0.375	0.016	0.019			
Q15_F~~Q15_F	0.469	0.018	0.021			
Q16_F~~Q16_F	0.756	0.034	0.034			
Q17_F~~Q17_F	0.192	0.012	0.013			
Q18_F~~Q18_F	0.255	0.015	0.015			
Q20_S~~Q20_S	0.302	0.025	0.023			
Q21_S~~Q21_S	0.765	0.035	0.036			
Q22_S~~Q22_S	0.374	0.023	0.023			
Q8_A~~Q8_A	0.759	0.037	0.035			
Q9_A~~Q9_A	0.549	0.029	0.027			
Q12_A~~Q12_A	0.645	0.037	0.035			
Q13_A~~Q13_A	0.433	0.027	0.021			
Q14_A~~Q14_A	0.492	0.026	0.026			
Q23_A~~Q23_A	0.670	0.034	0.032			
Q24_A~~Q24_A	0.711	0.033	0.033			
Q25_A~~Q25_A	0.731	0.035	0.034			
psycho~~physical	0.895	0.013	0.012			
psycho~~social	0.757	0.023	0.023			
psycho~~environment	0.706	0.025	0.023			
physical~~social	0.617	0.022	0.027			
physical~~environment	0.584	0.028	0.026			
social~~environment	0.553	0.031	0.031			
	Power (Not equal 0)	Std Est	Std Est	SD	Std Ave	SE
psycho=~Q5_P	1	0.636		0.017		0.020
psycho=~Q6_P	1	0.635		0.018		0.020
psycho=~Q7_P	1	0.637		0.016		0.020

psycho=~Q11_P	1	0.601	0.023	0.021
psycho=~Q19_P	1	0.821	0.010	0.012
psycho=~Q26_P	1	0.589	0.020	0.022
physical=~Q3_F	1	0.443	0.035	0.026
physical=~Q4_F	1	0.385	0.029	0.028
physical=~Q10_F	1	0.738	0.015	0.016
physical=~Q15_F	1	0.496	0.021	0.025
physical=~Q16_F	1	0.536	0.024	0.024
physical=~Q17_F	1	0.880	0.009	0.010
physical=~Q18_F	1	0.849	0.010	0.011
social=~Q20_S	1	0.787	0.018	0.019
social=~Q21_S	1	0.497	0.028	0.027
social=~Q22_S	1	0.719	0.020	0.021
environment=~Q8_A	1	0.455	0.026	0.028
environment=~Q9_A	1	0.568	0.024	0.024
environment=~Q12_A	1	0.682	0.022	0.020
environment=~Q13_A	1	0.602	0.022	0.023
environment=~Q14_A	1	0.678	0.020	0.021
environment=~Q23_A	1	0.515	0.023	0.026
environment=~Q24_A	1	0.485	0.027	0.027
environment=~Q25_A	1	0.478	0.028	0.027
Q3_F~~Q4_F	1	0.436	0.028	0.025
Q5_P~~Q5_P	1	0.595	0.022	0.026
Q6_P~~Q6_P	1	0.596	0.023	0.026
Q7_P~~Q7_P	1	0.594	0.020	0.026
Q11_P~~Q11_P	1	0.638	0.027	0.026
Q19_P~~Q19_P	1	0.325	0.017	0.020
Q26_P~~Q26_P	1	0.653	0.024	0.026
Q3_F~~Q3_F	1	0.802	0.030	0.023
Q4_F~~Q4_F	1	0.851	0.023	0.021
Q10_F~~Q10_F	1	0.456	0.022	0.023
Q15_F~~Q15_F	1	0.753	0.021	0.025
Q16_F~~Q16_F	1	0.712	0.025	0.025
Q17_F~~Q17_F	1	0.225	0.016	0.017
Q18_F~~Q18_F	1	0.279	0.016	0.018
Q20_S~~Q20_S	1	0.380	0.029	0.031
Q21_S~~Q21_S	1	0.752	0.028	0.027
Q22_S~~Q22_S	1	0.483	0.029	0.030
Q8_A~~Q8_A	1	0.792	0.023	0.025
Q9_A~~Q9_A	1	0.677	0.027	0.028
Q12_A~~Q12_A	1	0.534	0.029	0.028
Q13_A~~Q13_A	1	0.637	0.027	0.028
Q14_A~~Q14_A	1	0.539	0.027	0.028

Q23_A~~Q23_A	1	0.734	0.024	0.027
Q24_A~~Q24_A	1	0.764	0.026	0.026
Q25_A~~Q25_A	1	0.771	0.027	0.026
psycho~~physical	1	0.895	0.013	0.012
psycho~~social	1	0.757	0.023	0.023
psycho~~environment	1	0.706	0.025	0.023
physical~~social	1	0.617	0.022	0.027
physical~~environment	1	0.584	0.028	0.026
social~~environment	1	0.553	0.031	0.031

	Average Param	Average Bias	Coverage
psycho==Q5_P	0.529	-0.005	0.94
psycho==Q6_P	0.629	-0.001	0.96
psycho==Q7_P	0.546	0.001	1.00
psycho==Q11_P	0.597	0.000	0.90
psycho==Q19_P	0.778	0.000	0.98
psycho==Q26_P	0.543	0.002	0.98
physical==Q3_F	0.414	0.002	0.84
physical==Q4_F	0.387	0.000	0.90
physical==Q10_F	0.671	-0.002	0.94
physical==Q15_F	0.390	0.001	1.00
physical==Q16_F	0.552	0.001	0.96
physical==Q17_F	0.812	0.000	0.90
physical==Q18_F	0.813	0.000	0.98
social==Q20_S	0.697	0.005	0.98
social==Q21_S	0.500	0.001	0.92
social==Q22_S	0.633	-0.001	1.00
environment==Q8_A	0.450	-0.004	0.92
environment==Q9_A	0.513	-0.001	0.96
environment==Q12_A	0.740	0.010	0.96
environment==Q13_A	0.501	-0.004	0.92
environment==Q14_A	0.653	-0.005	0.96
environment==Q23_A	0.504	-0.012	0.94
environment==Q24_A	0.471	-0.002	0.94
environment==Q25_A	0.473	-0.007	0.92
Q3_F~~Q4_F	0.337	0.002	0.98
Q5_P~~Q5_P	0.397	0.004	0.96
Q6_P~~Q6_P	0.578	0.005	0.92
Q7_P~~Q7_P	0.442	-0.003	0.98
Q11_P~~Q11_P	0.626	0.002	0.98
Q19_P~~Q19_P	0.293	-0.001	0.98
Q26_P~~Q26_P	0.559	0.000	0.96
Q3_F~~Q3_F	0.700	0.002	0.90
Q4_F~~Q4_F	0.861	0.000	0.98

Q10_F~~Q10_F	0.379	-0.004	0.98
Q15_F~~Q15_F	0.475	-0.006	0.96
Q16_F~~Q16_F	0.765	-0.009	0.96
Q17_F~~Q17_F	0.191	0.001	0.96
Q18_F~~Q18_F	0.253	0.002	0.96
Q20_S~~Q20_S	0.305	-0.002	0.96
Q21_S~~Q21_S	0.757	0.008	0.94
Q22_S~~Q22_S	0.378	-0.004	0.98
Q8_A~~Q8_A	0.755	0.004	0.92
Q9_A~~Q9_A	0.556	-0.006	0.90
Q12_A~~Q12_A	0.652	-0.007	0.94
Q13_A~~Q13_A	0.428	0.005	0.88
Q14_A~~Q14_A	0.493	-0.001	0.96
Q23_A~~Q23_A	0.677	-0.007	0.94
Q24_A~~Q24_A	0.717	-0.005	0.96
Q25_A~~Q25_A	0.727	0.003	0.94
psycho~~physical	0.897	-0.002	0.92
psycho~~social	0.762	-0.005	0.96
psycho~~environment	0.702	0.003	0.98
physical~~social	0.620	-0.003	1.00
physical~~environment	0.581	0.003	0.88
social~~environment	0.552	0.001	0.94

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr
chisq	1.000	-0.178	-0.178	0.935	-0.891	-0.999	0.758
aic	-0.178	1.000	1.000	-0.154	0.142	0.179	-0.015
bic	-0.178	1.000	1.000	-0.154	0.142	0.179	-0.015
rmsea	0.935	-0.154	-0.154	1.000	-0.939	-0.936	0.685
cfi	-0.891	0.142	0.142	-0.939	1.000	0.898	-0.612
tli	-0.999	0.179	0.179	-0.936	0.898	1.000	-0.758
srmr	0.758	-0.015	-0.015	0.685	-0.612	-0.758	1.000

===== Replications =====

Number of replications = 50

Number of converged replications = 50

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

Source: [Model Comparison \(Null vs Alternative\)](#)

Alternative model simulation summary:RESULT OBJECT

Model Type

[1] "lavaan"

===== Fit Indices Cutoffs =====

Fit Indices	Alpha					Mean	SD
	0.1	0.05	0.01	0.001			
chisq	272.057	281.399	287.907	290.575	250.006	20.015	
aic	59544.727	59589.338	59669.433	59701.923	59259.129	214.603	
bic	59817.180	59861.790	59941.885	59974.375	59531.581	214.603	
rmsea	0.010	0.012	0.013	0.013	0.005	0.005	
cfi	0.998	0.997	0.997	0.997	0.999	0.001	
tli	0.998	0.997	0.996	0.996	1.000	0.002	
srmr	0.022	0.023	0.024	0.025	0.021	0.001	

===== Parameter Estimates and Standard Errors =====

	Estimate	Average	Estimate	SD	Average	SE
psycho=~Q5_P		0.920		0.021		0.024
psycho=~Q6_P		0.810		0.023		0.026
psycho=~Q7_P		0.940		0.020		0.024
psycho=~Q11_P		0.731		0.029		0.027
psycho=~Q19_P		0.747		0.027		0.027
psycho=~Q26_P		0.629		0.031		0.028
physical=~Q3_F		0.838		0.037		0.038
physical=~Q4_F		0.828		0.036		0.038
physical=~Q10_F		0.631		0.030		0.029
physical=~Q15_F		0.710		0.028		0.028
physical=~Q16_F		0.682		0.029		0.029
physical=~Q17_F		0.886		0.025		0.025
physical=~Q18_F		0.605		0.027		0.029
social=~Q20_S		0.937		0.024		0.024
social=~Q21_S		0.888		0.022		0.025
social=~Q22_S		0.780		0.031		0.027
environment=~Q8_A		0.372		0.034		0.031
environment=~Q9_A		0.488		0.032		0.030
environment=~Q12_A		0.793		0.031		0.027
environment=~Q13_A		0.812		0.029		0.026
environment=~Q14_A		0.772		0.028		0.027
environment=~Q23_A		0.752		0.028		0.027
environment=~Q24_A		0.641		0.031		0.029
environment=~Q25_A		0.732		0.029		0.028
Q3_F~~Q4_F		0.300		0.034		0.036
Q5_P~~Q5_P		0.149		0.011		0.010
Q6_P~~Q6_P		0.340		0.018		0.017
Q7_P~~Q7_P		0.122		0.010		0.009

Q11_P~~Q11_P	0.470	0.023	0.022
Q19_P~~Q19_P	0.442	0.021	0.021
Q26_P~~Q26_P	0.596	0.024	0.027
Q3_F~~Q3_F	1.005	0.045	0.048
Q4_F~~Q4_F	0.994	0.048	0.048
Q10_F~~Q10_F	0.600	0.029	0.029
Q15_F~~Q15_F	0.495	0.024	0.025
Q16_F~~Q16_F	0.543	0.031	0.027
Q17_F~~Q17_F	0.209	0.018	0.017
Q18_F~~Q18_F	0.628	0.040	0.030
Q20_S~~Q20_S	0.124	0.018	0.015
Q21_S~~Q21_S	0.207	0.015	0.016
Q22_S~~Q22_S	0.395	0.021	0.020
Q8_A~~Q8_A	0.864	0.037	0.038
Q9_A~~Q9_A	0.754	0.029	0.034
Q12_A~~Q12_A	0.375	0.020	0.020
Q13_A~~Q13_A	0.338	0.020	0.019
Q14_A~~Q14_A	0.411	0.026	0.022
Q23_A~~Q23_A	0.435	0.022	0.022
Q24_A~~Q24_A	0.581	0.029	0.028
Q25_A~~Q25_A	0.470	0.023	0.024
psycho~~physical	0.305	0.031	0.031
psycho~~social	0.293	0.032	0.030
psycho~~environment	0.293	0.031	0.031
physical~~social	0.299	0.033	0.031
physical~~environment	0.299	0.035	0.032
social~~environment	0.296	0.031	0.031

	Power (Not equal 0)	Std Est	Std Est	SD	Std Ave SE
psycho=~Q5_P	1	0.922	0.006		0.006
psycho=~Q6_P	1	0.811	0.010		0.012
psycho=~Q7_P	1	0.937	0.005		0.006
psycho=~Q11_P	1	0.729	0.016		0.015
psycho=~Q19_P	1	0.747	0.015		0.015
psycho=~Q26_P	1	0.631	0.021		0.019
physical=~Q3_F	1	0.641	0.019		0.020
physical=~Q4_F	1	0.639	0.019		0.021
physical=~Q10_F	1	0.631	0.021		0.021
physical=~Q15_F	1	0.710	0.018		0.018
physical=~Q16_F	1	0.679	0.021		0.019
physical=~Q17_F	1	0.889	0.010		0.011
physical=~Q18_F	1	0.606	0.025		0.022
social=~Q20_S	1	0.936	0.010		0.008
social=~Q21_S	1	0.890	0.008		0.010

social=~Q22_S	1	0.779	0.016	0.014
environment=~Q8_A	1	0.371	0.031	0.028
environment=~Q9_A	1	0.490	0.026	0.025
environment=~Q12_A	1	0.791	0.015	0.014
environment=~Q13_A	1	0.813	0.014	0.013
environment=~Q14_A	1	0.769	0.018	0.015
environment=~Q23_A	1	0.751	0.018	0.016
environment=~Q24_A	1	0.644	0.021	0.020
environment=~Q25_A	1	0.730	0.015	0.016
Q3_F~~Q4_F	1	0.300	0.028	0.030
Q5_P~~Q5_P	1	0.150	0.011	0.011
Q6_P~~Q6_P	1	0.342	0.017	0.019
Q7_P~~Q7_P	1	0.121	0.010	0.010
Q11_P~~Q11_P	1	0.468	0.023	0.022
Q19_P~~Q19_P	1	0.442	0.023	0.022
Q26_P~~Q26_P	1	0.601	0.026	0.025
Q3_F~~Q3_F	1	0.589	0.025	0.026
Q4_F~~Q4_F	1	0.592	0.024	0.026
Q10_F~~Q10_F	1	0.601	0.027	0.026
Q15_F~~Q15_F	1	0.496	0.026	0.025
Q16_F~~Q16_F	1	0.538	0.028	0.026
Q17_F~~Q17_F	1	0.210	0.018	0.019
Q18_F~~Q18_F	1	0.632	0.030	0.026
Q20_S~~Q20_S	1	0.123	0.018	0.016
Q21_S~~Q21_S	1	0.208	0.014	0.017
Q22_S~~Q22_S	1	0.394	0.025	0.021
Q8_A~~Q8_A	1	0.861	0.023	0.021
Q9_A~~Q9_A	1	0.760	0.025	0.025
Q12_A~~Q12_A	1	0.374	0.024	0.022
Q13_A~~Q13_A	1	0.340	0.022	0.021
Q14_A~~Q14_A	1	0.408	0.027	0.023
Q23_A~~Q23_A	1	0.435	0.026	0.023
Q24_A~~Q24_A	1	0.585	0.027	0.026
Q25_A~~Q25_A	1	0.468	0.022	0.024
psycho~~physical	1	0.305	0.031	0.031
psycho~~social	1	0.293	0.032	0.030
psycho~~environment	1	0.293	0.031	0.031
physical~~social	1	0.299	0.033	0.031
physical~~environment	1	0.299	0.035	0.032
social~~environment	1	0.296	0.031	0.031
Average Param Average Bias Coverage				
psycho=~Q5_P	0.92	0.000	0.98	
psycho=~Q6_P	0.81	0.000	0.98	

psycho=~Q7_P	0.94	0.000	0.98
psycho=~Q11_P	0.73	0.001	0.92
psycho=~Q19_P	0.75	-0.003	0.96
psycho=~Q26_P	0.63	-0.001	0.96
physical=~Q3_F	0.84	-0.002	0.92
physical=~Q4_F	0.83	-0.002	0.98
physical=~Q10_F	0.63	0.001	0.94
physical=~Q15_F	0.71	0.000	0.96
physical=~Q16_F	0.68	0.002	0.92
physical=~Q17_F	0.89	-0.004	0.94
physical=~Q18_F	0.61	-0.005	0.96
social=~Q20_S	0.94	-0.003	0.94
social=~Q21_S	0.89	-0.002	0.96
social=~Q22_S	0.78	0.000	0.96
environment=~Q8_A	0.37	0.002	0.94
environment=~Q9_A	0.49	-0.002	0.94
environment=~Q12_A	0.79	0.003	0.94
environment=~Q13_A	0.81	0.002	0.96
environment=~Q14_A	0.77	0.002	0.96
environment=~Q23_A	0.75	0.002	0.94
environment=~Q24_A	0.64	0.001	0.94
environment=~Q25_A	0.73	0.002	0.96
Q3_F~~Q4_F	0.30	0.000	0.96
Q5_P~~Q5_P	0.15	-0.001	0.96
Q6_P~~Q6_P	0.34	0.000	0.94
Q7_P~~Q7_P	0.12	0.002	0.94
Q11_P~~Q11_P	0.47	0.000	0.92
Q19_P~~Q19_P	0.44	0.002	0.98
Q26_P~~Q26_P	0.60	-0.004	0.92
Q3_F~~Q3_F	1.00	0.005	0.98
Q4_F~~Q4_F	1.00	-0.006	0.94
Q10_F~~Q10_F	0.60	0.000	0.98
Q15_F~~Q15_F	0.50	-0.005	0.94
Q16_F~~Q16_F	0.54	0.003	0.92
Q17_F~~Q17_F	0.21	-0.001	0.94
Q18_F~~Q18_F	0.63	-0.002	0.84
Q20_S~~Q20_S	0.12	0.004	0.90
Q21_S~~Q21_S	0.21	-0.003	0.92
Q22_S~~Q22_S	0.39	0.005	0.94
Q8_A~~Q8_A	0.86	0.004	0.98
Q9_A~~Q9_A	0.76	-0.006	0.96
Q12_A~~Q12_A	0.38	-0.005	0.94
Q13_A~~Q13_A	0.34	-0.002	0.96

Q14_A~~Q14_A	0.41	0.001	0.88
Q23_A~~Q23_A	0.44	-0.005	0.96
Q24_A~~Q24_A	0.59	-0.009	0.90
Q25_A~~Q25_A	0.47	0.000	0.96
psycho~~physical	0.30	0.005	0.94
psycho~~social	0.30	-0.007	0.94
psycho~~environment	0.30	-0.007	0.96
physical~~social	0.30	-0.001	0.92
physical~~environment	0.30	-0.001	0.90
social~~environment	0.30	-0.004	0.94

===== Correlation between Fit Indices =====

	chisq	aic	bic	rmsea	cfi	tli	srmr
chisq	1.000	-0.107	-0.107	0.919	-0.893	-1.000	0.530
aic	-0.107	1.000	1.000	-0.126	0.098	0.106	-0.005
bic	-0.107	1.000	1.000	-0.126	0.098	0.106	-0.005
rmsea	0.919	-0.126	-0.126	1.000	-0.961	-0.919	0.515
cfi	-0.893	0.098	0.098	-0.961	1.000	0.893	-0.490
tli	-1.000	0.106	0.106	-0.919	0.893	1.000	-0.532
srmr	0.530	-0.005	-0.005	0.515	-0.490	-0.532	1.000

===== Replications =====

Number of replications = 50

Number of converged replications = 50

Number of nonconverged replications:

1. Nonconvergent Results = 0
2. Nonconvergent results from multiple imputation = 0
3. At least one SE were negative or NA = 0
4. Nonpositive-definite latent or observed (residual) covariance matrix (e.g., Heywood case or linear dependency) = 0

Source: [Model Comparison \(Null vs Alternative\)](#)

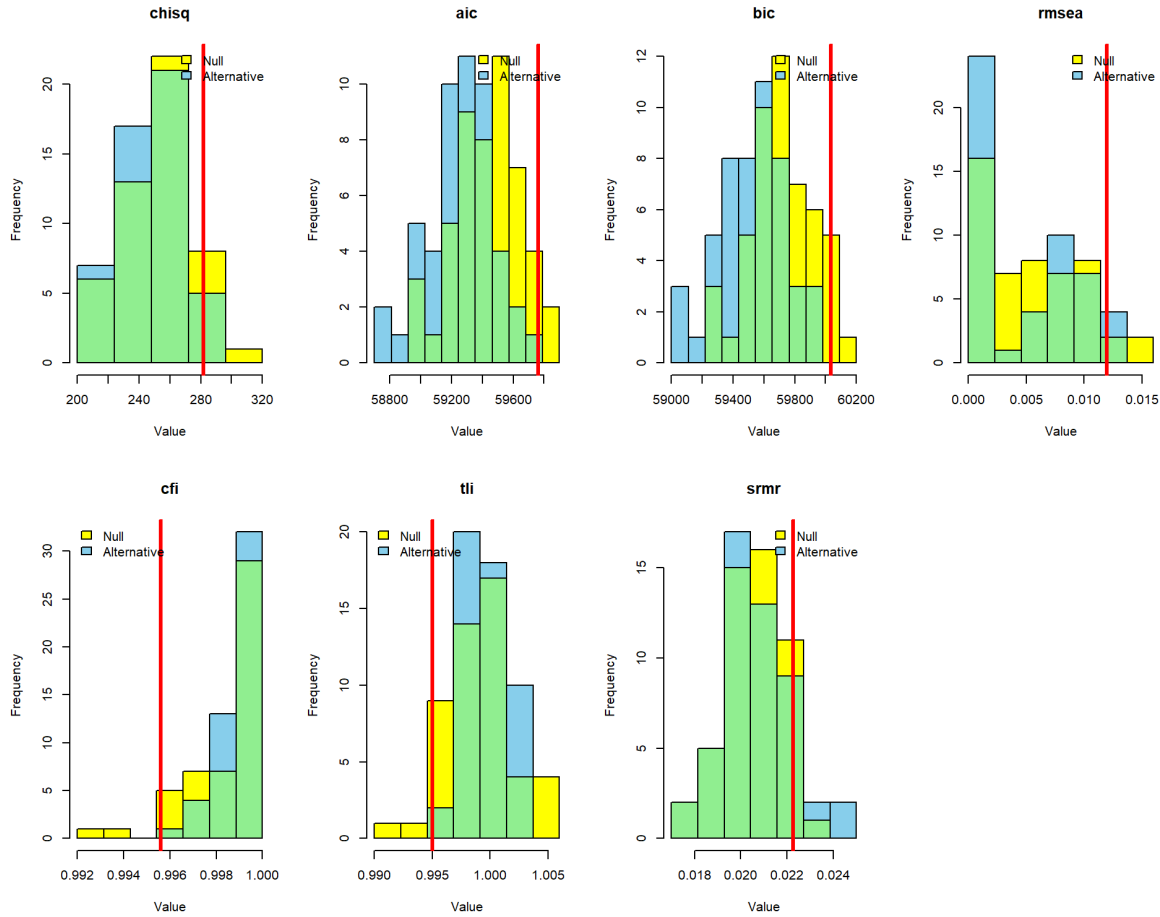


Figure 5: Fit index distributions — Null (empirical) vs. Alternative (population)

Source: [Model Comparison \(Null vs Alternative\)](#)

Table 6: Post hoc power to distinguish empirical from population model (N = 1047)

Fit Index	Null Cutoff (= .05)	Power
RMSEA	From cutoff_null above	From getPowerFit above
CFI	From cutoff_null above	From getPowerFit above
TLI	From cutoff_null above	From getPowerFit above
SRMR	From cutoff_null above	From getPowerFit above

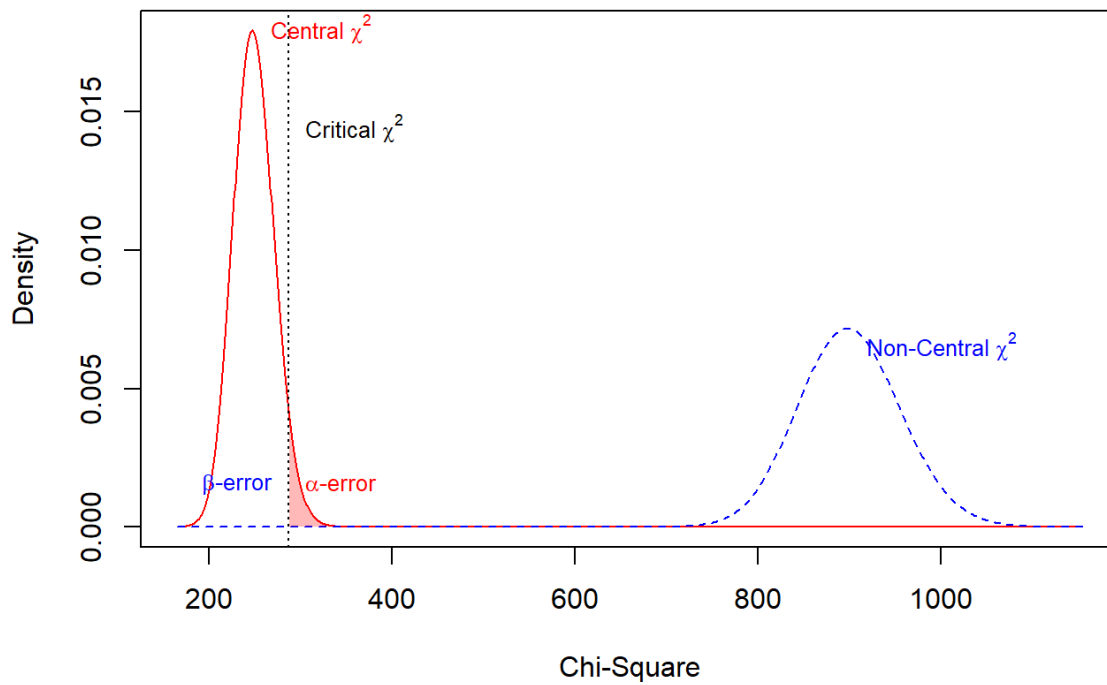
Source: [Model Comparison \(Null vs Alternative\)](#)

7 Post Hoc: Achieved Power

For the empirical WHOQOL-BREF dataset, $N = 1,047$. What is our achieved power if true RMSEA is 0.05?

semPower: Post hoc power analysis

F0	0.622500
RMSEA	0.050000
Mc	0.732531
df	249
Num Observations	1047
NCP	651.1350
Critical Chi-Square	286.8077
Alpha	0.050000
Beta	4.771426e-48
Power (1 - Beta)	> 0.9999
Implied Alpha/Beta Ratio	1.047905e+46



Source: [Post Hoc Power Analysis](#)

Achieved power (`semTools`): 1

Source: [Post Hoc Power Analysis](#)

8 When to Use What: Analytical vs. Simulation

Feature	Analytical (<code>semPower</code> / <code>semTools</code>)	Simulation (<code>simsem</code>)
Speed	Instantaneous	Minutes to hours
Handles non-normality effectively	No	Yes
Handles missing data mechanism	No	Yes

Feature	Analytical (<code>semPower</code> / <code>semTools</code>)	Simulation (<code>simsem</code>)
Parameter-level power targeting	Very Limited	Comprehensive
Coverage / relative bias diagnostics	No	Yes
Primary Use-Case	Grant planning, quick rough parameterization bounds	Pre-registration confirmation, sensitivity checks, final reporting

Best Practice Recommendation: Define theoretical bounds mapping an expected scenario utilizing fast analytical approximations. Validate robust tolerance to observed/anticipated data flaws (e.g., missingness, skew, differing distributions) iteratively through explicit simulation runs before finalizing sample specifications.

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